

DATA SCIENCE METHODOLOGY

MODULE 3

Overview

- Analytics for Data Science
- Examples of Data Analytics
- Data Analytics Lifecycle
- Data Discovery
- Data Preparation
- Model Planning
- Model Building
- Communicate Results

Analytics for Data Science

- **Analytics is the process of examining data to extract meaningful insights, trends, and patterns that support decision-making.**
- In Data Science, analytics is used **before, during, and after** building models.

Types of Analytics

- Descriptive Analytics
- Diagnostic Analytics
- Predictive Analytics
- Prescriptive Analytics

Descriptive Analytics - “What happened?”

Objective: Summarize past data.

Examples:

- Average marks of students last semester
- Number of dropouts per department
- Website traffic this month

Tools:

- Excel
- SQL
- Power BI/Tableau

Outputs:

- charts, graphs, reports, dashboards

Diagnostic Analytics - “Why did it happen?”

Objective: Find reasons behind outcomes.

Examples:

- Dropout increased because attendance went down
- Sales dropped when price increased
- Server failures increased during peak load

Techniques:

- Correlation analysis
- Drill-down reports
- Root cause analysis

Predictive Analytics - “What will happen next?”

Objective: Use machine learning to forecast outcomes.

Examples:

- Predict student dropout risk
- Predict future sales
- Predict hospital readmission
- Detect fraud

Techniques:

- Regression
- Classification
- Time-series forecasting
- Clustering

Tools:

- Python, R, ML libraries

Predictive Analytics - “What will happen next?”

Objective: Recommend actions based on predictions.

Examples:

- Suggest intervention plans for at-risk students
- Recommend discounts for customers likely to leave
- Optimize staffing schedules

Techniques:

- Optimization
- Decision rules
- Reinforcement learning

Outcome:

- Turns data insights into **actions**.

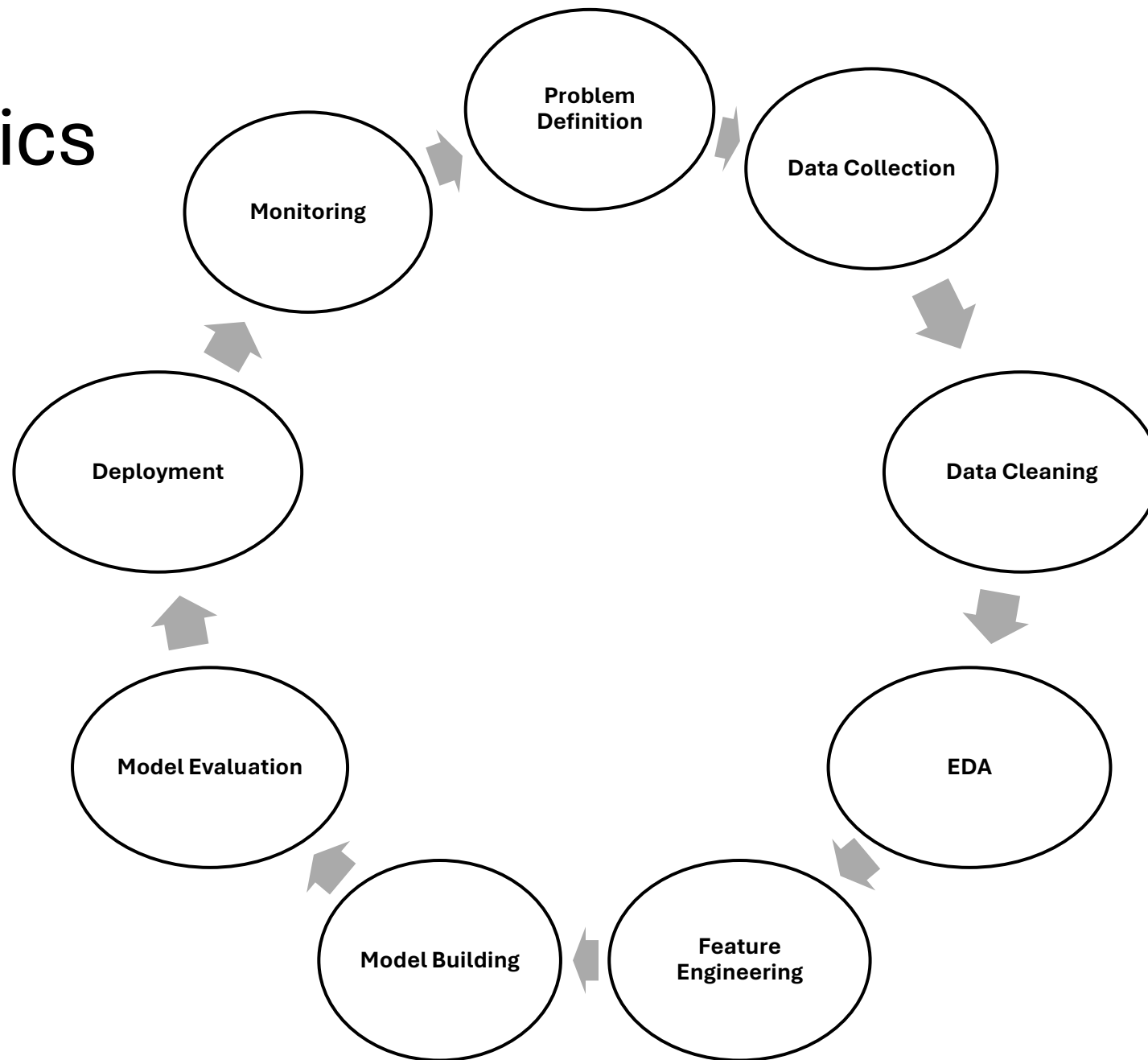
How Analytics Fits in Data Science?

Data Science = **Data + Analytics + Machine Learning + Decision Support**

Analytics helps in:

- understanding data
 - discovering issues
 - selecting right features
 - evaluating model results
 - communicating insights
- Without analytics, **data science models become blind guesses.**

Data Analytics Lifecycle



Data Analytics Lifecycle

Step	One-Line Definition
Problem Definition	Understand what needs to be solved.
Data Collection	Gather all required raw data.
Data Cleaning	Fix and prepare data for analysis.
EDA	Explore and visualize data to derive insights.
Feature Engineering	Create meaningful variables for modeling.
Model Building	Train ML models using the prepared dataset.
Model Evaluation	Check if the model performs well.
Deployment	Move the model into production.
Monitoring	Track model performance and update as needed.

Problem Definition

Questions answered:

- What problem are we solving?
 - What decision will this analysis support?
 - What does success look like?
-
- **Examples:** Predict customer churn, detect fraud, classify medical images, etc.

Data Collection

Gather data from multiple sources:

- Databases
- APIs
- Web scraping
- Sensors / IoT
- Surveys
- Public datasets
- **Key output:** Raw dataset.

Data Cleaning (Preprocessing)

This takes 60–80% of the time.

Tasks include:

- Handling missing values
- Removing duplicates
- Fixing inconsistent formatting
- Treating outliers
- Encoding categorical variables
- Normalization/standardisation
- **Goal:** Convert raw data into usable form.

Exploratory Data Analysis (EDA)

Understand patterns, trends and relationships.

Tasks:

- Descriptive statistics
 - Data visualization
 - Correlations
 - Feature distributions
-
- **Outputs:** Insights + hypotheses + feature ideas.

Feature Engineering

Improve model quality by transforming data meaningfully.

Includes:

- Creating new features
 - Feature selection
 - Dimensionality reduction (PCA)
 - Domain-specific transformations
-
- **Goal:** Make data more informative for the model.

Model Building (Machine Learning)

Choose and train models such as:

- Regression
- Classification algorithms
- Decision trees
- Random Forest, XGBoost
- Deep learning models

Tasks:

- Train/test split
- Hyperparameter tuning
- Cross-validation

Model Evaluation

- Metrics depend on the problem type:

Problem Type	Metrics
Classification	Accuracy, F1-score, ROC-AUC
Regression	RMSE, MAE, R^2
Clustering	Silhouette score
Deep Learning	Loss, Precision-Recall

Model Deployment

Make the model accessible for real-world use.

Methods:

- REST API
- Cloud deployment (AWS, GCP, Azure)
- Edge devices
- Mobile/Web apps

Monitoring & Maintenance (MLOps)

Monitor:

- Model drift
- Data drift
- Performance degradation

Update the model when new data arrives.

- This brings the workflow back to step 1 → Continuous cycle.

Use Case

“How can a university accurately identify students who are at high risk of dropping out using available academic, behavioural and wellbeing data so that timely interventions can be implemented to improve student retention?”

Problem Definition

- **Develop a data-driven system** that analyses student information from multiple sources such as academic performance, attendance, LMS activity, stress levels, engagement scores and financial data to **predict the likelihood of dropout early in the semester.**
- System must:
 - Detect patterns associated with dropout risk
 - Classify students into Low, Medium or High risk categories
 - Provide actionable insights for academic counsellors
 - Enable proactive support (mentoring, tutoring, counselling)
 - Reduce dropout rates and improve overall student success
- **Build an intelligent, early-warning prediction model** that identifies vulnerable students before failure or withdrawal occurs.

Data Collection

Data Source Category	Specific Systems / Tools	Example Data Collected (Features)
1. Databases	Student Information System (SIS)	Demographics (age, gender), program enrolled, academic history, CGPA, arrears/backlogs
	Learning Management System (LMS)	Login frequency, time spent, assignment submissions, quiz/exam scores
	Fee / Finance System	Fee status, pending dues, scholarship/waiver details, payment delays
2. APIs	Attendance System API	Attendance %, absentee patterns, late entry/early exit events
	Learning Apps / MOOC API	Course progress, video watch time, completion rate, activity streaks
3. Web Scraping	University/College Website Portals	Notices viewed, course materials accessed, academic calendar interaction
	Discussion Forums / Course Groups	Student engagement in discussions, number of posts/comments, doubts asked

Data Collection

Data Source Category	Specific Systems / Tools	Example Data Collected (Features)
4. Sensors / IoT	Smart Classroom Devices	Entry/exit logs, in-class presence, frequency of class visits
	Wi-Fi / Beacon Access Logs	Time spent on campus, hotspot switching pattern, building-level footfall
5. Surveys	Student Satisfaction Surveys	Campus experience score, faculty rating, classroom experience
	Mental Health / Stress / Workload Surveys	Stress level, burnout risk, workload score, emotional well-being
6. Public Datasets	Government Employment & Education Portals	Local unemployment rate, job opportunities in student's region
	Socio-economic Indicators	Household income index, literacy rate, area development score, urban/rural classification

Raw Data Sheet

Student_ID	Hostel_Resident (Yes/No)
Age	Parent_Income_Level (Low/Medium/High)
Gender	Mental_Health_Score (1–10)
Attendance_%	Engagement_Score (1–10)
CGPA	Part_Time_Work (Yes/No)
Arrears_Count	Distance_From_Campus_KM
LMS_Login_Per_Week	WiFi_Hours_On_Campus
Assignment_Submission_Rate_%	Survey_Stress_Level (Low/Med/High)
Fee_Due (Yes/No)	-

How to Identify Important Features?

- Five complementary approaches:
 1. Domain Knowledge Filtering (Education Logic)
 2. Statistical Correlation Analysis
 3. Feature Importance Using Simple ML Models
 4. Variance Threshold (Remove Low-Information Features)
 5. PCA (Dimensionality Reduction)

Domain Knowledge Filtering (Education Logic)

Strong Features	Why It Matters	Weak Features	Why It Matters
Attendance %	Direct indicator of risk	Student_ID	Identifier only
CGPA	Low academic performance → high dropout	Department	Weak unless dropout varies by dept
Arrears/Backlogs	Very strong predictor	Gender	Usually negligible
LMS Logins	Digital engagement = academic engagement	Hostel Resident	Weak predictor unless strong social factors
Assignment Submission %	Discipline & study habits	Parent Income	Moderate but indirect
Stress Level	Emotional strain affects performance	-	-
Mental Health Score	Strong early warning	-	-
Engagement Score	Motivation indicator	-	-
WiFi Hours	Time spent on campus	-	-

Statistical Correlation Analysis

- Use Pearson correlation to identify how strongly each feature relates to the target (risk).

Features Considered	Correlation Score
Attendance ↔ Risk	-0.72 (strong)
CGPA ↔ Risk	-0.81 (very strong)
Stress ↔ Risk	+0.75 (strong)
LMS Logins ↔ Risk	-0.55 (moderate)
WiFi Hours ↔ Risk	-0.48 (moderate)
Parent Income ↔ Risk	-0.10 (weak)
Gender ↔ Risk	0.02 (no correlation)

Retain - absolute correlation > 0.3

Drop - absolute correlation < 0.1

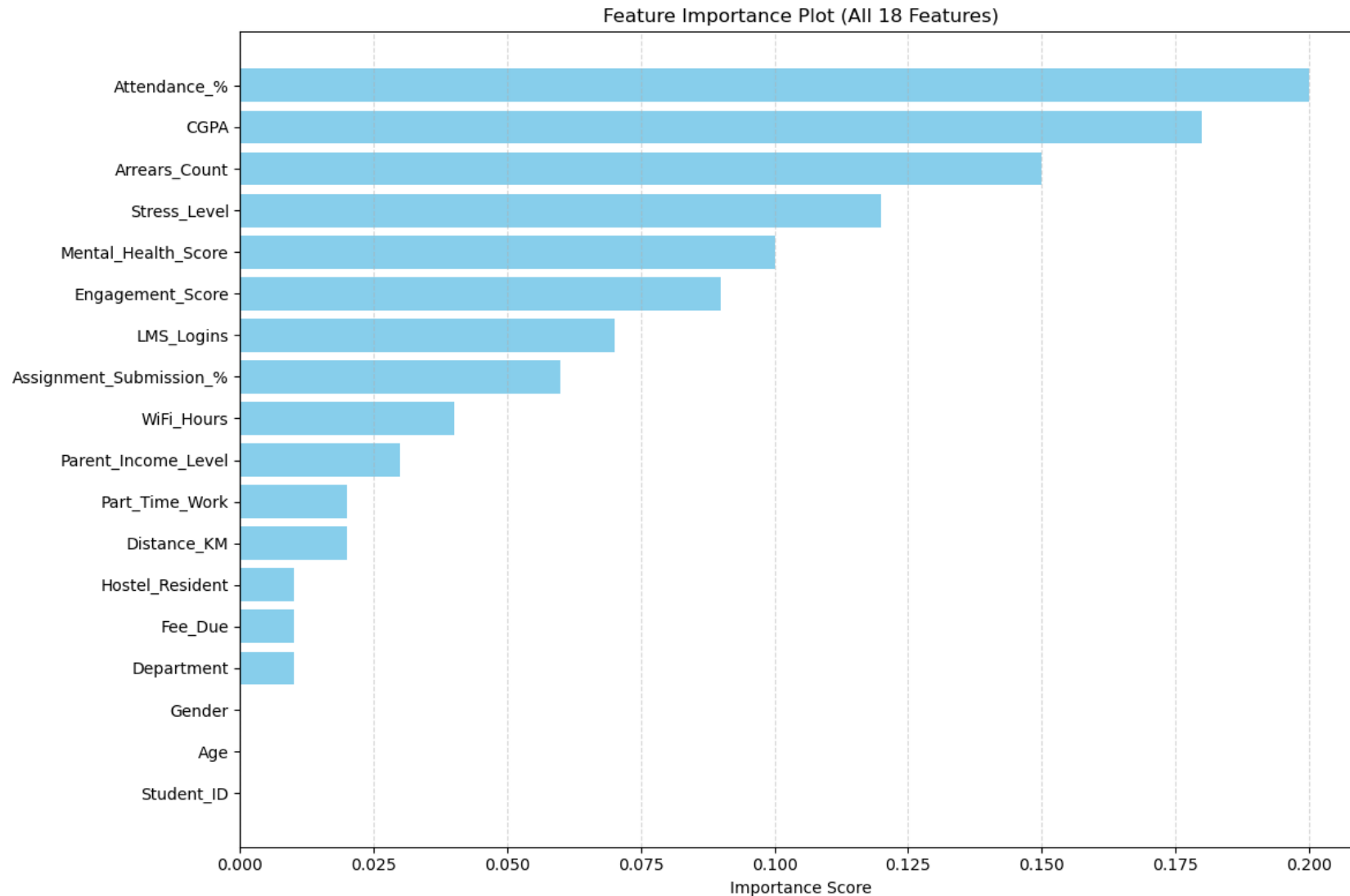
Feature Importance Using Simple ML Models

- Two *simple models* quickly reveal important features:
 - **Decision Tree**
 - **Random Forest**
- They automatically assign importance scores.

Feature	Importance
CGPA	0.28
Attendance	0.22
Stress Level	0.18
Mental Health	0.14
LMS Logins	0.10
Assignment Submission	0.06
WiFi Hours	0.02
Gender	0.00

Remove features with importance < 0.02

Feature Importance Using Simple ML Models



Variance Threshold (Remove Low-Information Features)

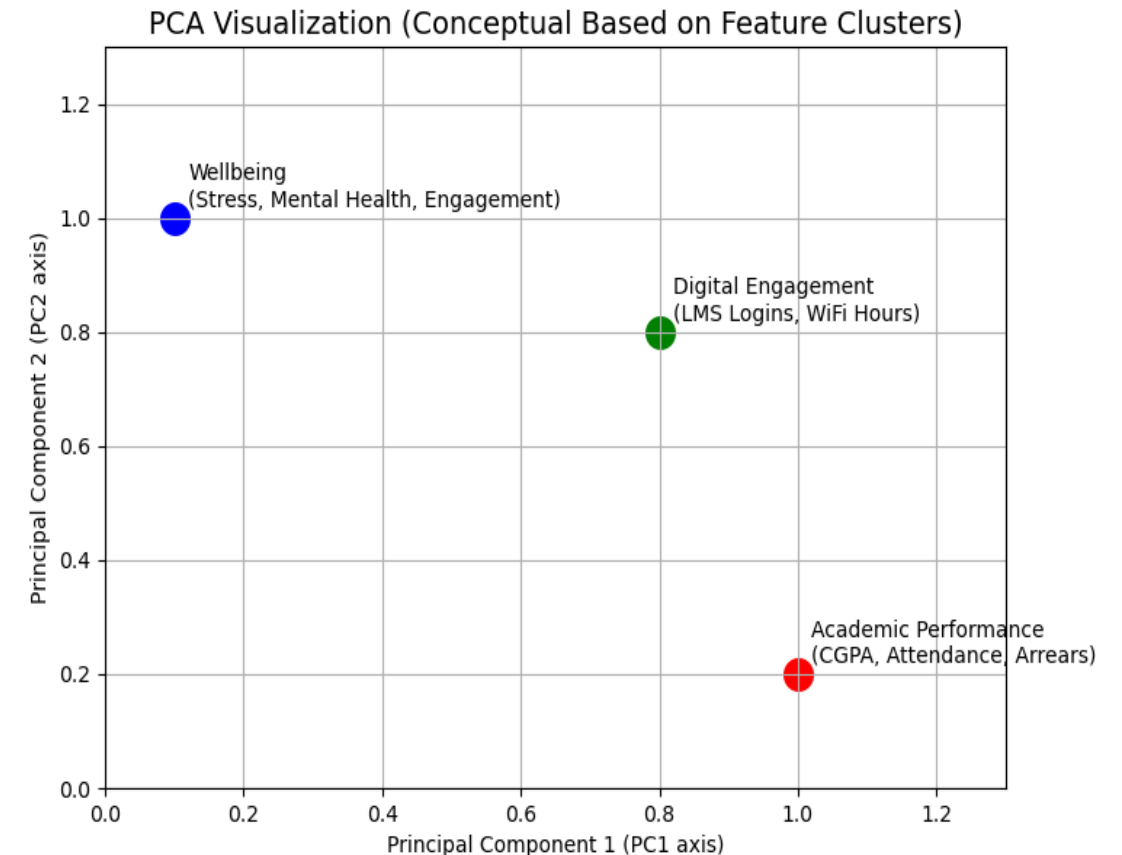
- Features that almost never change → useless

Feature	Values	Variance	Keep?
Student_ID	All unique	Low	✗ Remove (identifier, not useful)
Gender	Mostly “F”	Very low	✗ Remove
Hostel_Resident	[1,1,1,0,1,1,1]	Very low	✗ Remove
Attendance	[48,55,60,70,88,92]	High	✓ Keep
CGPA	[5.6,6.1,7.3,8.4,9.1,9.0]	High	✓ Keep
Stress_Level_Num	[0,0,1,2,2,0]	Medium	✓ Keep

PCA (Dimensionality Reduction)

- PCA is not for feature *selection* → helps identify clusters of related features.

Principal Component	High-Loading Features	Meaning
PC1	CGPA, Attendance, Arrears	Academic performance
PC2	Stress, Mental Health, Engagement	Wellbeing
PC3	LMS Logins, WiFi Hours	Digital engagement



Final Features

Keep These (Strong Predictors)

1. Attendance %
2. CGPA
3. Arrears Count
4. LMS Login Per Week
5. Assignment Submission Rate
6. Stress Level
7. Mental Health Score
8. Engagement Score
9. WiFi Hours on Campus
10. Parent Income (optional)

Maybe Keep (Depends on dataset)

11. Distance From Campus
12. Part-Time Work
13. Hostel Resident

Remove (Weak or Irrelevant)

14. Student_ID
15. Gender
16. Department
17. Fee Due
18. Survey text fields / redundant features

Data Cleaning Process

1. Remove Duplicate Records
2. Handle Missing Values
3. Fix Inconsistent Formats
4. Convert Categorical Data to Numeric Codes
5. Fix Outliers
6. Normalize Numerical Values
7. Detect Skewed Data
8. Feature Consistency Across Systems
9. Final Clean Dataset Structure

Data Cleaning Process

Cleaning Step	Issue	Raw Example	Cleaning Method	Cleaned Example
1. Remove Duplicates	Same record appears twice	S010, 82%, 7.3 (duplicate row)	Keep first or latest instance	One entry kept
2. Handle Missing Values	Missing attendance	Attendance = None	Replace with median (≈ 75)	75
	Missing CGPA	""	Replace with dept-average	7.2
	Missing LMS logins	blank	Replace with 0	0
3. Fix Inconsistent Formats	Percent stored as text	"85%"	Strip % and convert	85
	CGPA format incorrect	"07.50"	Convert to float	7.5
	Department case mismatch	"cse"	Standardize	"CSE"
	Gender mismatch	"Female"	Convert	"F"

Data Cleaning Process

Cleaning Step	Issue	Raw Example	Cleaning Method	Cleaned Example
4. Encode Categorical Values	Stress levels are strings	Low, Med, High	Encode as numbers	0, 1, 2
	Gender strings	"M","F"	Encode	0,1
	FeeDue yes/no	Yes	Encode	1
	Department categories	CSE, ECE, MECH	Map to numbers	0,1,2
5. Remove Outliers	Impossible CGPA	12.5	Clip to max 10	10
	Negative attendance	-20	Set to valid min	0
	WiFi Hours too high	120	Replace with median (≈ 20)	20
	LMS Logins too high	200	Cap at 25	25
6. Normalize Values	Attendance range too large	75	Scale 0–1	0.75
	CGPA range	8.4	Divide by 10	0.84
	LMS Logins	15	Divide by 25	0.60

Data Cleaning Process

Cleaning Step	Issue	Raw Example	Cleaning Method	Cleaned Example
7. Handle Skewed Data	Many small LMS values	0,1,2,25	Log/binning	Low, Med, High
	WiFi Hours long tail	5,6,40	Cap extreme	40 → 30
8. Merge Multi-source Data	LMS entry missing	Student exists only in SIS	Set LMS = 0	0
	Missing attendance week	API blank	Weighted average	Filled value
9. Produce ML-Ready Final Row	Mixed types, ranges, errors	Raw unclean row	After all cleaning	Attend=0.58, CGPA=0.61, Stress=2, LMS=3, WiFi=20, Gender=0, Dept=1

Data Cleaning - Issues

- Duplicate student records
- Missing values in **Age, Attendance, LMS logins, Assignment %, Mental Health, Engagement**
- Inconsistent **Gender** formats (M, F, Male, female, m, FEMALE...)
- Inconsistent **Department** values (CSE/cse/Mech/Bio/IT/it...)
- Messy **CGPA** (strings, comma format, values > 10)
- Mixed-case categorical fields (Yes/NO/no/Y/N etc.)
- Irregular **Survey Stress Level** values (Low/Med/High + variations)
- Random missing values to simulate sensor/API failures

Sample Dataset

Student	Attendance	CGPA	Fee_Due	Department
S1	85	8.2	No	CSE
S2	72	NaN	Yes	CSE
S3	NaN	7.1	No	ECE
S4	65	6.5	NaN	EEE
S5	90	9.1	No	CSE

NaN – Missing Values

Handling Missing Values

- Row Deletion
- Feature (Column) Deletion

Row Deletion

- Remove rows with missing values.
- Rows removed → S2, S3, S4

Student	Attendance	CGPA
S1	85	8.2
S5	90	9.1

- Lost **60% of data** - risky unless missing data is small.

Feature (Column) Deletion

Fee_Due has missing values?

| Fee_Due column: | No, Yes, NaN, No, No |

- If >50% missing we delete column:

Student	Attendance	CGPA	Department
S1	85	8.2	CSE
S2	72	NaN	CSE
S3	NaN	7.1	ECE
S4	65	6.5	EEE
S5	90	9.1	CSE

Imputation Methods

- **Imputation** is the process of replacing missing data with substituted values.
- Types:
 - **Mean / Median Imputation**
 - **Mode Imputation**
 - **Constant / Zero Imputation**
 - **Multivariate Imputation**
 - KNN Imputation
 - Regression Imputation
 - **Transformation**
 - Smoothing
 - Aggregation
 - **Normalization**
 - Min-max scaling
 - Z-score Standardization
 - Decimal Scaling

Mean / Median Imputation (Univariate)

CGPA values: 8.2, NaN, 7.1, 6.5, 9.1

Mean = 7.72

Median = 7.65

Student	Attendance	CGPA	Department
S1	85	8.2	CSE
S2	72	7.72 / 7.65	CSE
S3	NaN	7.1	ECE
S4	65	6.5	EEE
S5	90	9.1	CSE

Mode Imputation (Categorical)

- Fee_Due values: No, Yes, NaN, No, No
- Mode = **No (Most occurring value)**
- Replace missing: Nan

Student	Attendance	CGPA	Fee_Due	Department
S1	85	8.2	No	CSE
S2	72	NaN	Yes	CSE
S3	NaN	7.1	No	ECE
S4	65	6.5	No	EEE
S5	90	9.1	No	CSE

Constant / Zero Imputation

- Attendance missing at S3:
- Replace with 0 (or “Unknown” depending on meaning)

Student	Attendance	CGPA	Fee_Due	Department
S1	85	8.2	No	CSE
S2	72	NaN	Yes	CSE
S3	0	7.1	No	ECE
S4	65	6.5	No	EEE
S5	90	9.1	No	CSE

Multivariate Imputation - KNN

- **KNN Imputation**
- For CGPA at S2, look at students similar in attendance:
- S2 Attendance = 72
- Closest are S1 (85) and S4 (65) and S3 (0)
- Average their CGPA ≈ 7.3

Student	Attendance	CGPA	Fee_Due	Department
S1	85	8.2	No	CSE
S2	72	NaN 7.3	Yes	CSE
S3	0	7.1	No	ECE
S4	65	6.5	No	EEE
S5	90	9.1	No	CSE

Multivariate Imputation - KNN

- **Step 1: Compute distances from S2 (Attendance = 72)**
- Use simple absolute difference: $\text{distance} = |x_{\text{other}} - x_{\text{S2}}|$

Student	Attendance	Distance from 72
S1	85	13
S3	0	72
S4	65	7
S5	90	18

Multivariate Imputation - KNN

Step 2: Pick the k nearest neighbours

If we choose **k = 3**, the **3 smallest distances** are:

- S4: distance 7 (Attendance 65, CGPA 6.5)
- S1: distance 13 (Attendance 85, CGPA 8.2)
- S5: distance 18 (Attendance 90, CGPA 9.1)

Student	Attendance	CGPA	Fee_Due	Department
S1	85	8.2	No	CSE
S2	72	NaN	Yes	CSE
S3	0	7.1	No	ECE
S4	65	6.5	No	EEE
S5	90	9.1	No	CSE

- So the *correct* 3 nearest neighbours are **S4, S1, S5**
(S3 with attendance 0 is actually *far* from 72, so it should **not** be chosen).

Multivariate Imputation - KNN

- **Step 3: Average their CGPA values**

$$\text{CGPA}_{S_2} = \frac{6.5 + 8.2 + 9.1}{3} = \frac{23.8}{3} = 7.933 \dots \approx \boxed{7.93}$$

- So, with proper KNN (k=3, nearest in attendance), **imputed CGPA for S2 ≈ 7.93**

Multivariate Imputation - Regression

- Regression Imputation
- Build a regression model

Student	Attendance	CGPA	Fee_Due	Department
S1	85	8.2	No	CSE
S2	72	NaN	Yes	CSE
S3	0	7.1	No	ECE
S4	65	6.5	No	EEE
S5	90	9.1	No	CSE

Multivariate Imputation - Regression

- Use rows with CGPA available:

Attendance (x)	CGPA (y)
85	8.2
0	7.1
65	6.5
90	9.1

Multivariate Imputation - Regression

- Step 1: Compute Means

$$\bar{x} = \frac{85 + 0 + 65 + 90}{4} = \frac{240}{4} = 60$$

$$\bar{y} = \frac{8.2 + 7.1 + 6.5 + 9.1}{4} = \frac{30.9}{4} = 7.725$$

Multivariate Imputation - Regression

- Step 2: Compute slope 'b'

$$b = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

Multivariate Imputation - Regression

- Step 2: Compute slope 'b'

x	y	(x-x')	(y-y')	Product
85	8.2	25	0.475	11.875
0	7.1	-60	-0.625	37.5
65	6.5	5	-1.225	-6.125
90	9.1	30	1.375	41.25

$$\sum (x - \bar{x})(y - \bar{y}) = 11.875 + 37.5 - 6.125 + 41.25 = 84.5$$

Multivariate Imputation - Regression

- Step 2: Compute slope 'b'

x	(x-x')	Square
85	25	625
0	-60	3600
65	5	25
90	30	900

$$b = \frac{84.5}{5150} \approx 0.0164$$

$$\sum (x - \bar{x})^2 = 625 + 3600 + 25 + 900 = 5150$$

Multivariate Imputation - Regression

- Step 3: Compute intercept 'a'

$$a = \bar{y} - b\bar{x} = 7.725 - (0.0164 \times 60) = 7.725 - 0.984 = 6.741$$

So regression equation becomes:

$$\hat{y} = 6.741 + 0.0164x$$

Multivariate Imputation - Regression

- Step 4: Predict CGPA when Attendance = 72

$$\hat{y} = 6.741 + 0.0164(72) = 6.741 + 1.1808 = \boxed{7.9218}$$

Rounded:

$$\boxed{\text{Predicted CGPA} \approx 7.92}$$

Based on existing students, a student with **72% attendance** is expected to have approximately **CGPA $\approx$ 7.9**.

Transformation

- Transformation is the process of converting raw data into a better, more useful form by smoothing, scaling, combining, or reshaping it.

We transform data so it becomes:

- easier to analyse
- less noisy
- suitable for machine learning models
- more meaningful for interpretation

Transformation – Smoothing

Smoothing reduces random fluctuations (noise) in the data.

Data often contains spikes caused by:

- human errors
- sensor glitches
- unusual rare events
- Smoothing makes the **trend clearer**.

Transformation – Smoothing

Daily attendance: 70, 72, 68, **90**, 69, 71 (The spike **90** may just be noise or exception)

Use a **3-day moving average**: $MA_3 = \frac{x_{t-2} + x_{t-1} + x_t}{3}$

- First average (days 1-3): $MA_3 = \frac{70+72+68}{3} = \frac{210}{3} = 70$
- Second average (days 2-4): $MA_3 = \frac{72+68+90}{3} = \frac{230}{3} = 76.67 \approx 77$
- Third average (days 3-5): $MA_3 = \frac{68+90+69}{3} = \frac{227}{3} = 75.67 \approx 76$
- Fourth average (days 4-6): $MA_3 = \frac{90+69+71}{3} = \frac{230}{3} = 76.67 \approx 77$

Transformation – Aggregation

- Aggregation combines multiple records into a single summary value.
- Reduces dataset size
- Removes day-to-day random variation
- Simplifies analysis
- Makes results more understandable

Transformation – Aggregation

- Instead of tracking attendance **every day**, compute an **average for the entire month**.
- Example: 80, 85, 82, 79, 78, . . . (more days)
- Assume 30 daily values, represented symbolically:

$$\text{Monthly Avg} = \frac{\sum_{i=1}^{30} \text{Attendance}}{30}$$

- Suppose the total sum across 30 days is: Total = 2430
- Then:

$$\text{Monthly Mean} = \frac{2430}{30} = 81\%$$

Exploratory Data Analysis

EDA helps us **understand patterns, trends, and hidden structures** in the student dataset before modelling.

We examine:

- Descriptive statistics
- Data visualizations
- Correlations
- Feature distributions

Normalization

When do we normalize?

Normalize when models rely on:

- **distance** (KNN, K-Means)
- **gradients** (Neural networks)
- **margins** (SVM, logistic regression)
- Not strictly needed for Decision Trees / Random Forests (but still fine).

Normalization

Method	Scales To	Best For	Notes
Min-Max	$0 \rightarrow 1$	Neural nets, KNN, SVM	Sensitive to outliers
Z-Score	mean 0, std 1	Most ML models	Handles outliers better
Decimal Scaling	depends on k	Large numeric IDs	Rarely used in ML

Normalization – Min-Max (0-1)

$$\text{Formula: } x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

Example:

Student	Attendance
S1	40
S2	80
S3	60

For **S3 (60)**: Min – 40 & Max - 80

$$x' = \frac{60 - 40}{80 - 40} = \frac{20}{40} = 0.5$$

- Keeps everything between **0 and 1**
- Preserves shape but rescales

Normalization – Z-Score (mean=0, std=1)

$$\text{Formula: } z = \frac{x - \mu}{\sigma}$$

Where: μ = mean, σ = standard deviation

Example: Mean Attendance = 60 & Std Dev = 16

For S1 (40):

$$z = \frac{40 - 60}{16} = -1.25$$

- Values become centered around 0
- Good → data looks “normal” (bell-shaped)

Normalization – Decimal Scaling

Formula: $x' = \frac{x}{10^k}$ Where, choose $k = [-1, 1]$

- Example: Given $x=4500$, choose $k=3$

$$x' = \frac{4500}{10^3} = 4.5$$

- We effectively moved the decimal: $4500 \rightarrow 450.0 \rightarrow 45.0 \rightarrow \mathbf{4.5}$

Normalization – Decimal Scaling

Student	Total_Fees_Paid (₹)
S1	4500
S2	12000
S3	8000

Max = 12000 → 5 digits → choose $k = 4$ (or 5)

Normalization – Decimal Scaling

- If $k = 4 \rightarrow x' = \frac{x}{10^4}$

Student	Original	Normalized (x')
S1	4500	0.45
S2	12000	1.2
S3	8000	0.8

Normalization – Decimal Scaling

- If we want all values < 1 , choose $k = 5 \rightarrow x' = \frac{x}{10^5}$

Student	Original	Normalized (x')
S1	4500	0.045
S2	12000	0.12
S3	8000	0.08

Exploratory Data Analysis

- **Exploratory Data Analysis (EDA)** is the process of exploring and analysing a dataset to understand its structure, patterns, relationships, and problems - before applying machine learning or modelling.
- EDA helps to:
 - detect missing values & errors
 - find outliers
 - identify patterns and trends
 - understand distributions
 - choose the right preprocessing steps
 - decide which algorithms may work
 - avoid misleading conclusions

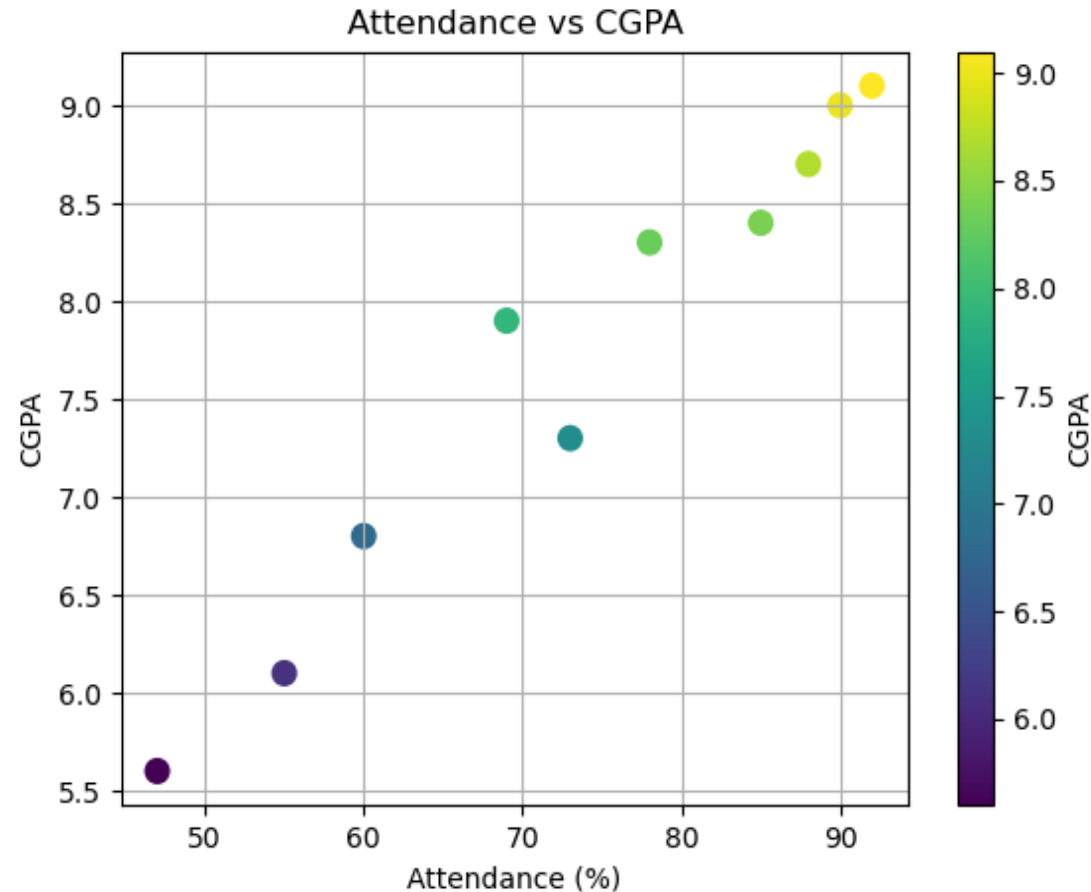
EDA – Descriptive Statistics

- These statistics summarize the overall behaviour of students.

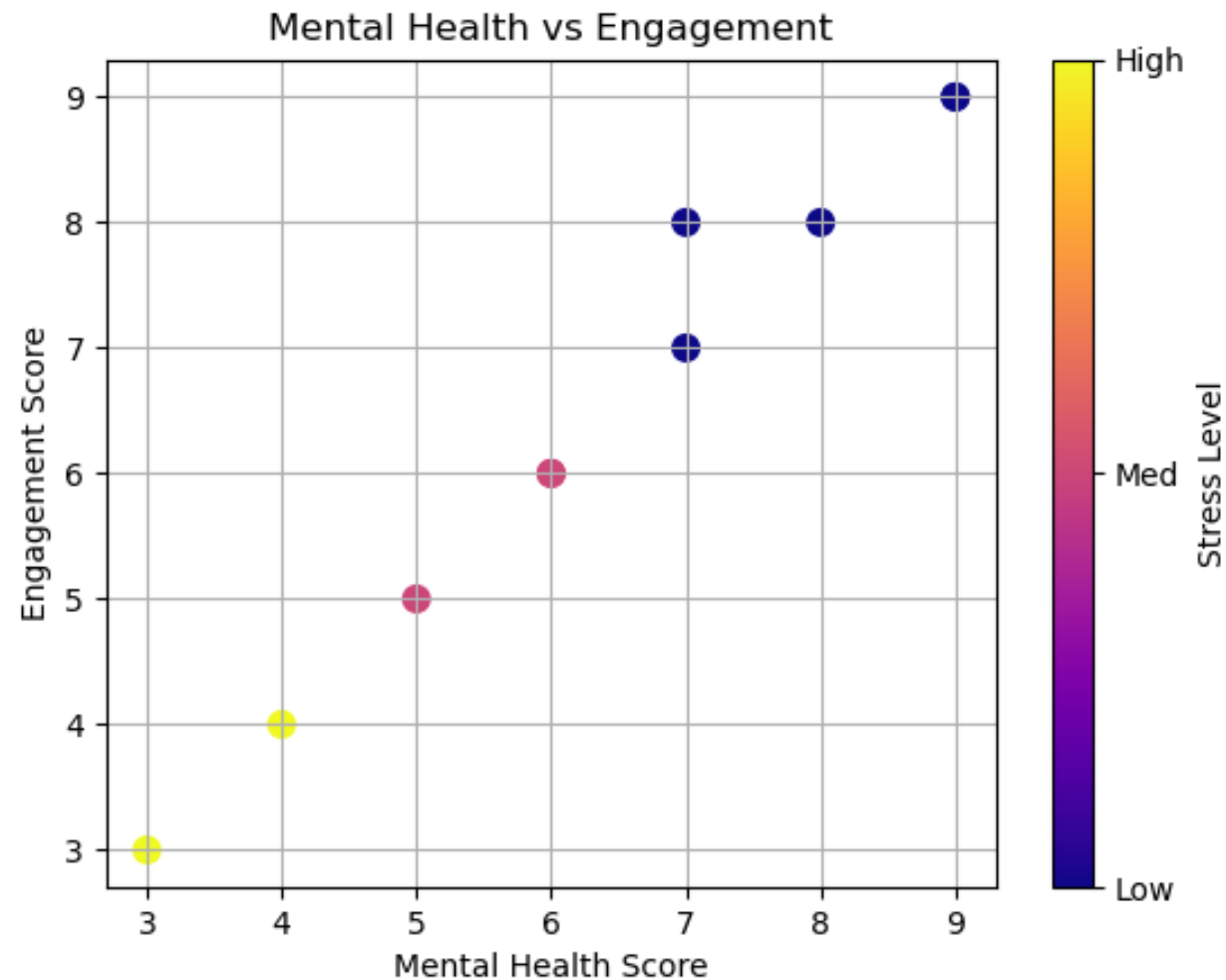
Feature	Mean	Min	Max	Notes
Attendance (%)	73.6	47	95	Many students have borderline attendance
CGPA	7.42	5.13	9.70	Wide range → strong academic variation
LMS Logins/Week	12.0	2	24	Low logins → possible academic disengagement
WiFi Hours	22.4	5	41	On-campus time varies greatly
Mental Health	6.4	2	10	Several students show stress risk
Engagement Score	6.4	2	10	Mirrors mental health trends

EDA – Data Visualization

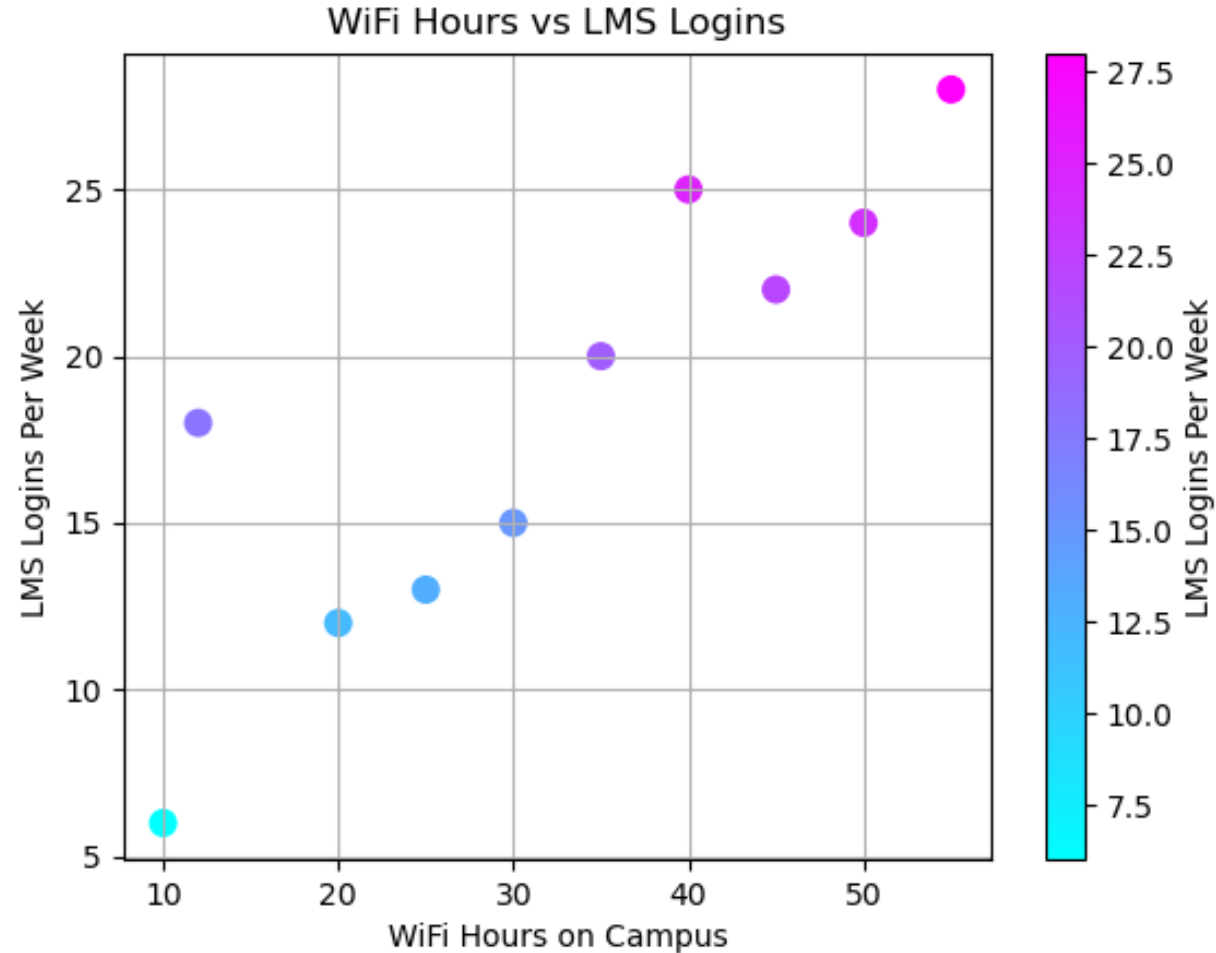
- These plots reveal relationships visually



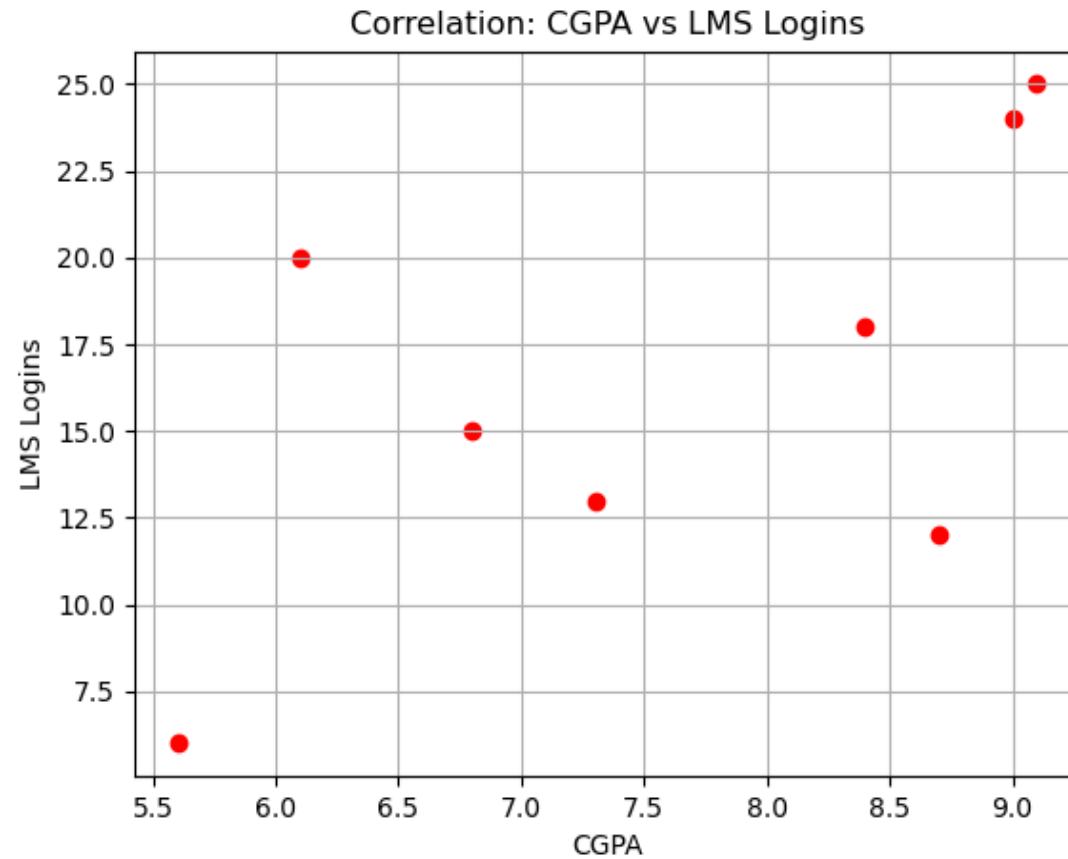
EDA – Data Visualization



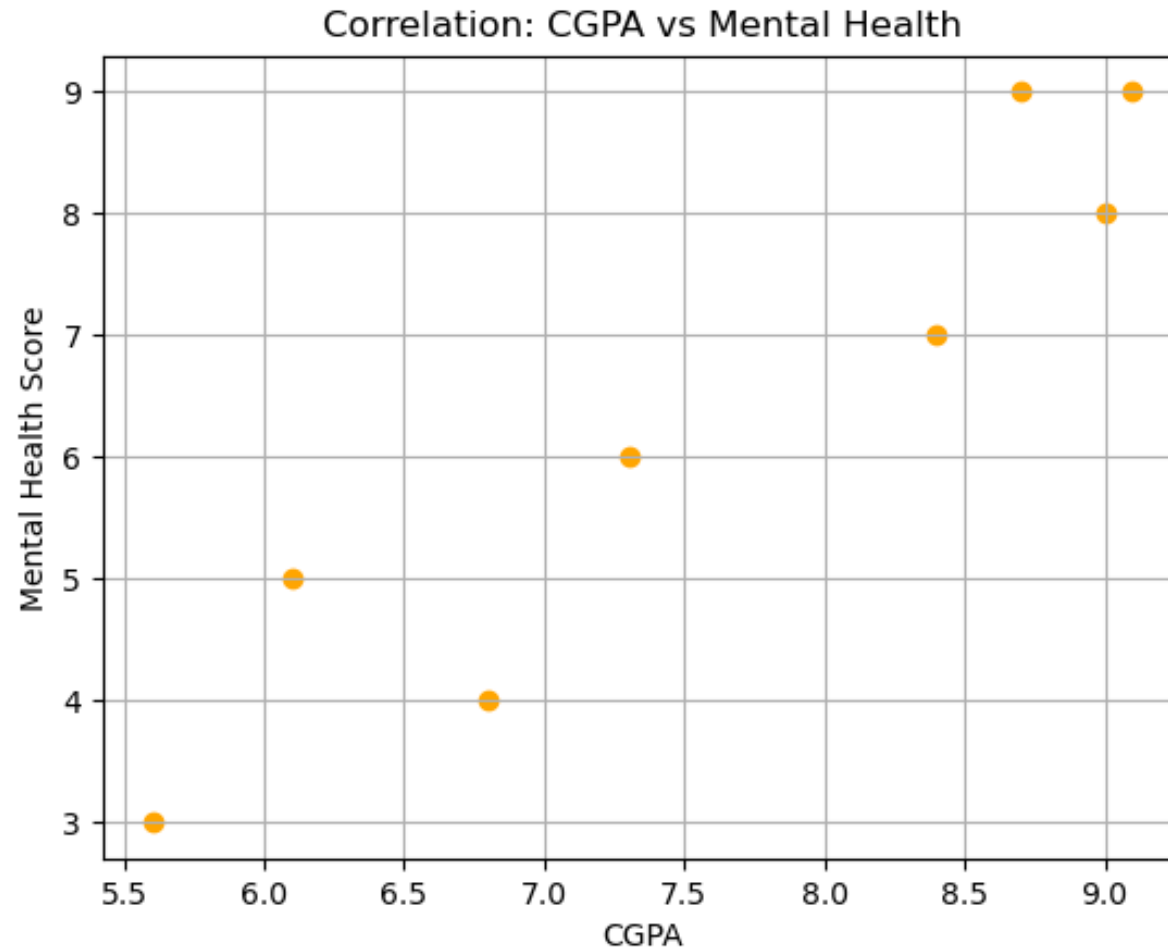
EDA – Data Visualization



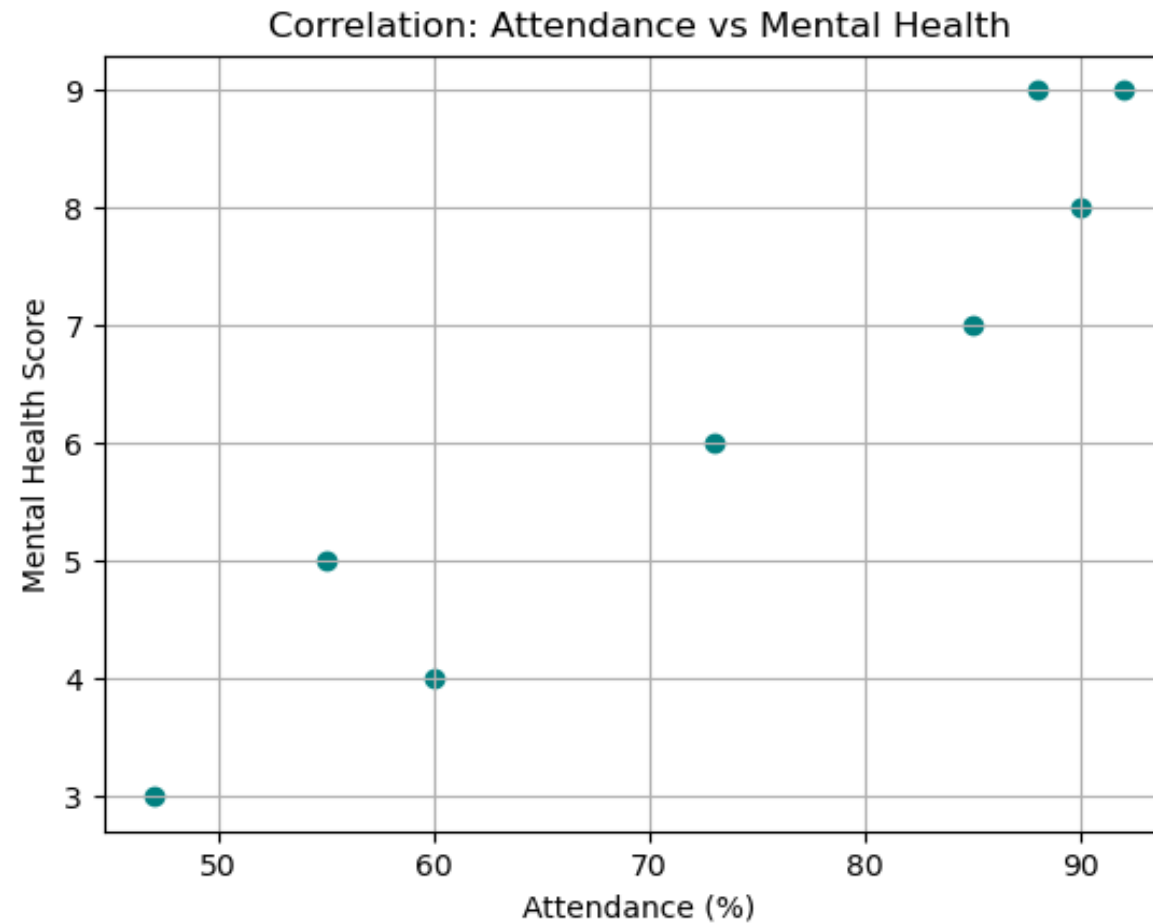
EDA – Data Visualization



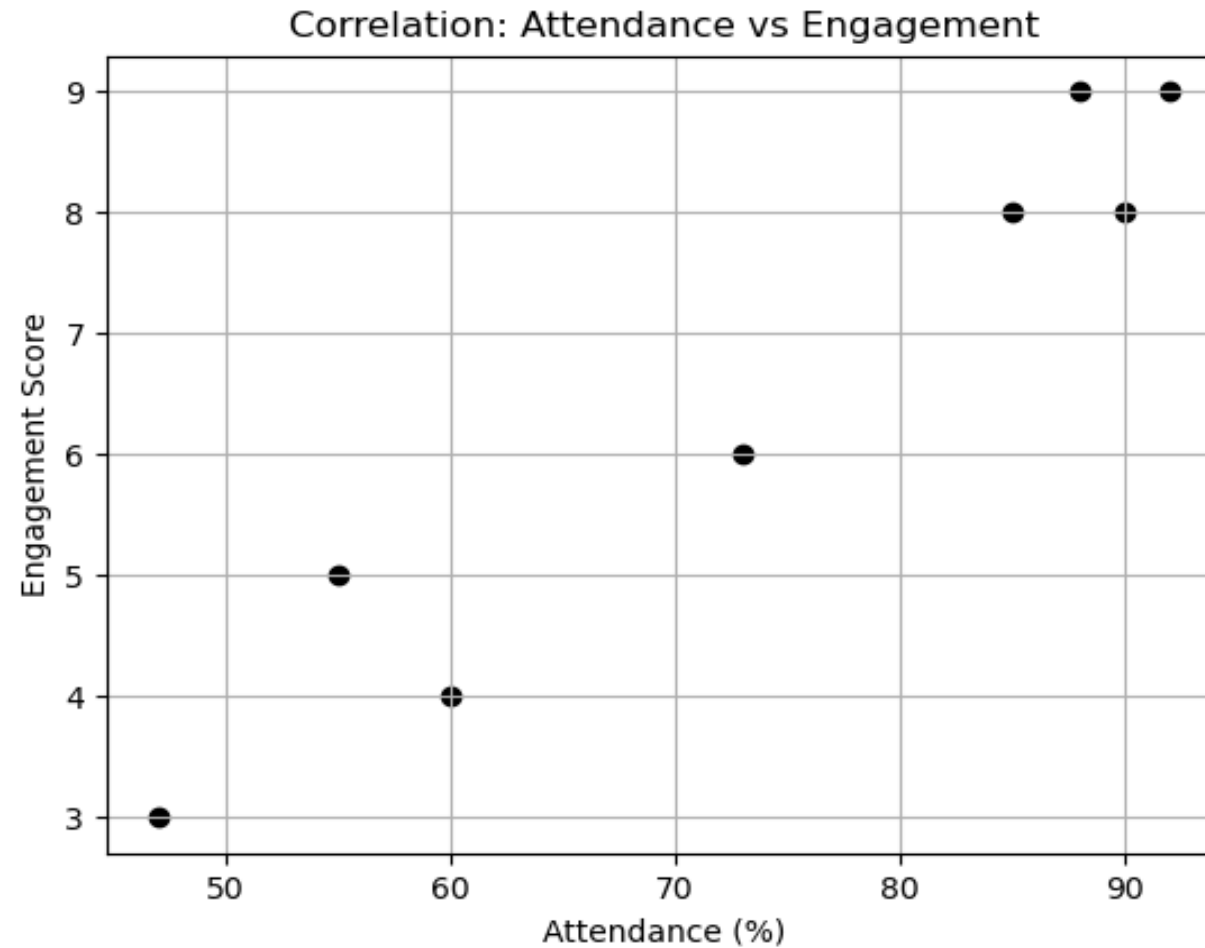
EDA – Data Visualization



EDA – Data Visualization

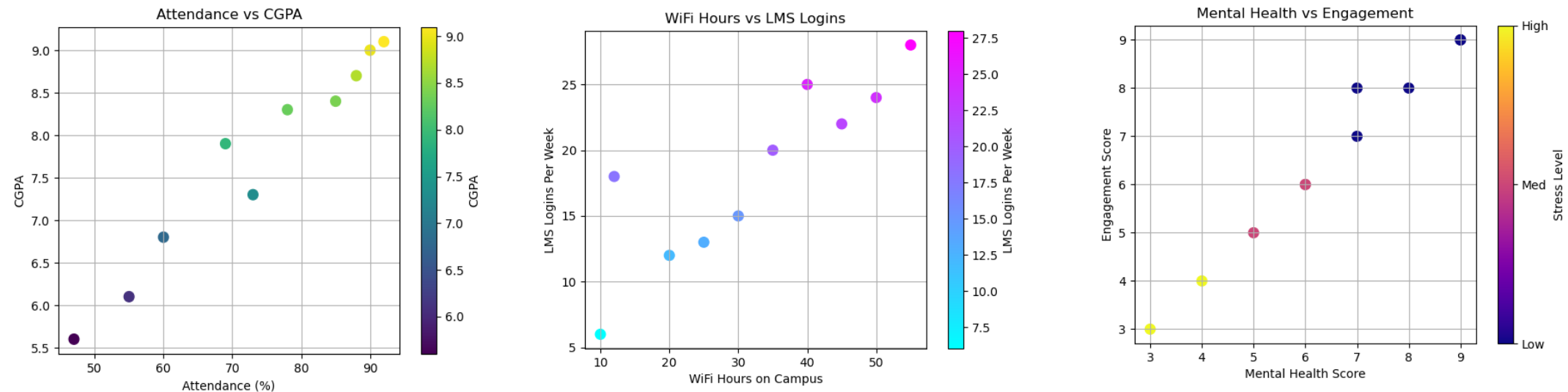


EDA – Data Visualization



EDA – Correlations

- Correlations quantify how strongly two factors move together.

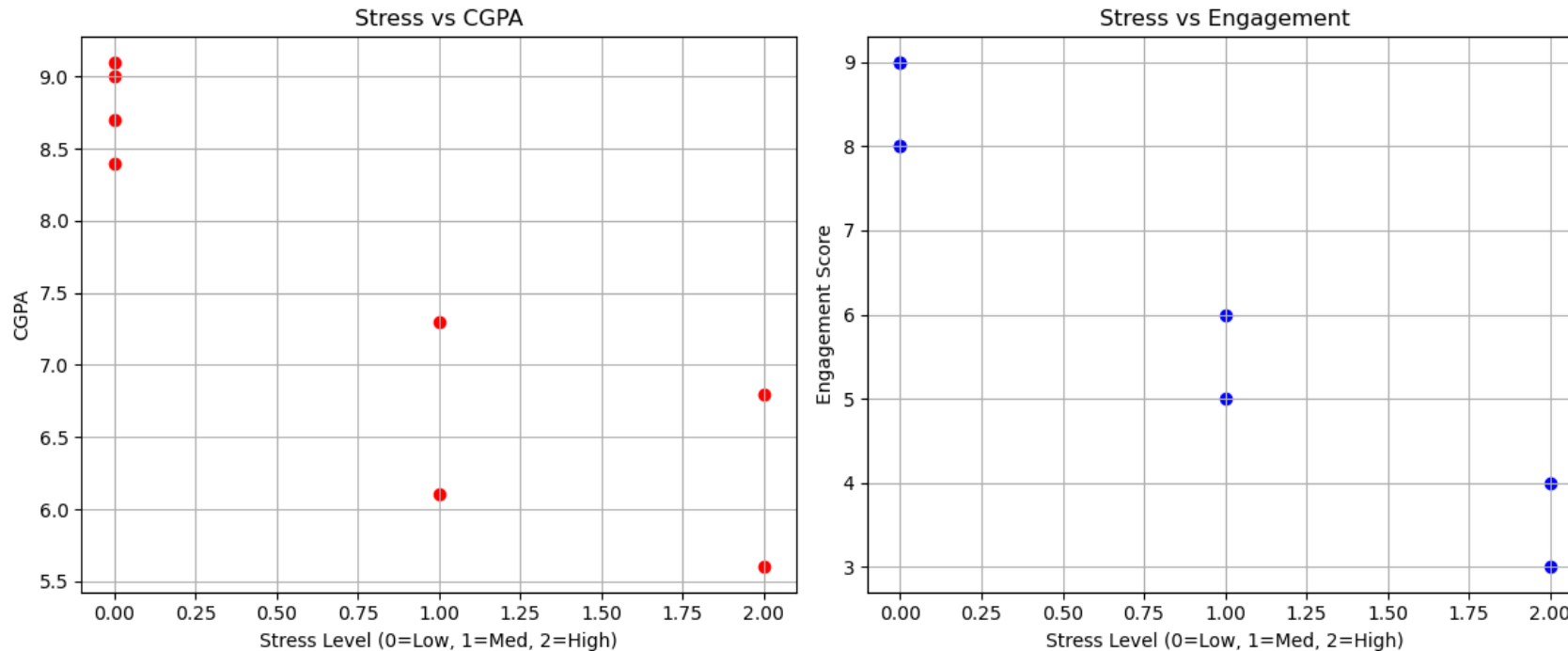


High Positive Correlations: When one variable increases, the other also increases.

- Attendance $\leftrightarrow$ CGPA
- LMS Logins $\leftrightarrow$ WiFi Hours
- Engagement $\leftrightarrow$ Mental Health

EDA – Correlations

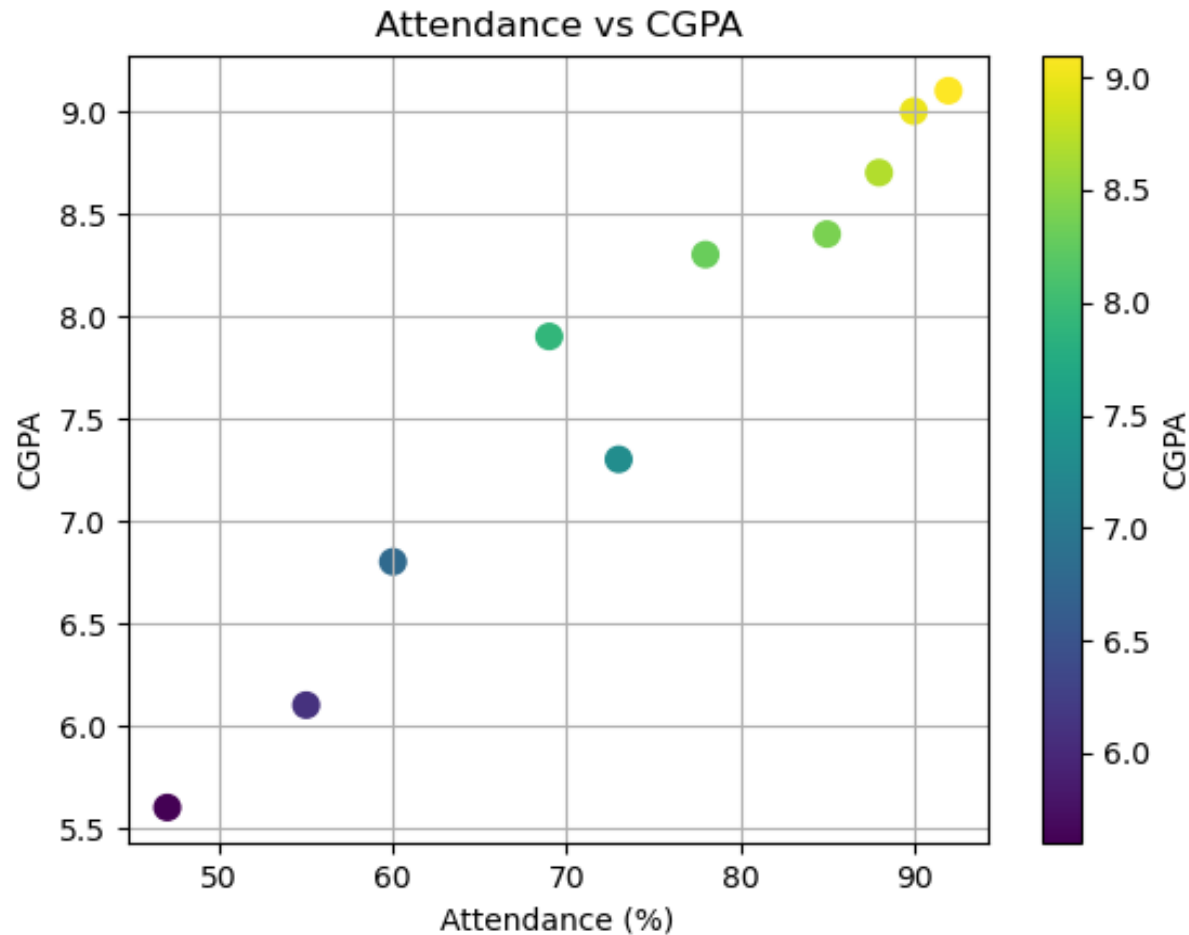
- Correlations quantify how strongly two factors move together.



High Negative Correlations: When one variable increases, the other decreases.

- Stress $\leftrightarrow$ CGPA
- Stress $\leftrightarrow$ Engagement

EDA - Correlation



Attendance (x)	CGPA (y)
48	5.6
55	6.1
60	6.8
65	6.9
70	7.9
75	7.3
80	8.3
85	8.6
88	8.4
90	9.0
92	9.1

EDA - Correlation

- Calculate Means

$$\bar{x} = \frac{\sum x}{n}$$

$$\bar{y} = \frac{\sum y}{n}$$

Attendance Mean:

$$\bar{x} = \frac{48 + 55 + 60 + 65 + 70 + 75 + 80 + 85 + 88 + 90 + 92}{11}$$

$$\bar{x} = \frac{808}{11} = 73.45$$

CGPA Mean:

$$\bar{y} = \frac{5.6 + 6.1 + 6.8 + 6.9 + 7.9 + 7.3 + 8.3 + 8.6 + 8.4 + 9.0 + 9.1}{11}$$

$$\bar{y} = \frac{84.0}{11} = 7.64$$

EDA - Correlation

- Compute Deviations

x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$	$(x - \bar{x})^2$	$(y - \bar{y})^2$
48	5.6	-25.45	-2.04	51.31	647.7	4.16
55	6.1	-18.45	-1.54	28.43	340.4	2.37
60	6.8	-13.45	-0.84	11.30	181.0	0.71
65	6.9	-8.45	-0.74	6.25	71.4	0.55
70	7.9	-3.45	0.26	-0.90	11.9	0.07
75	7.3	1.55	-0.34	-0.53	2.40	0.11
80	8.3	6.55	0.66	4.32	42.9	0.44
85	8.6	11.55	0.96	11.09	133.4	0.92
88	8.4	14.55	0.76	11.06	211.7	0.58
90	9.0	16.55	1.36	22.51	274.0	1.85
92	9.1	18.55	1.46	27.11	344.2	2.13

EDA - Correlation

- Compute Pearson Correlation r

Formula:

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \cdot \sum (y - \bar{y})^2}}$$

Compute sums:

$$\sum (x - \bar{x})(y - \bar{y}) = 171.95$$

$$\sum (x - \bar{x})^2 = 2261.0$$

$$\sum (y - \bar{y})^2 = 13.49$$

EDA - Correlation

- Plug into formula

$$r = \frac{171.95}{\sqrt{2261.0 \times 13.49}}$$

Calculate denominator:

$$\sqrt{2261.0 \times 13.49} = \sqrt{30513.89} = 174.64$$

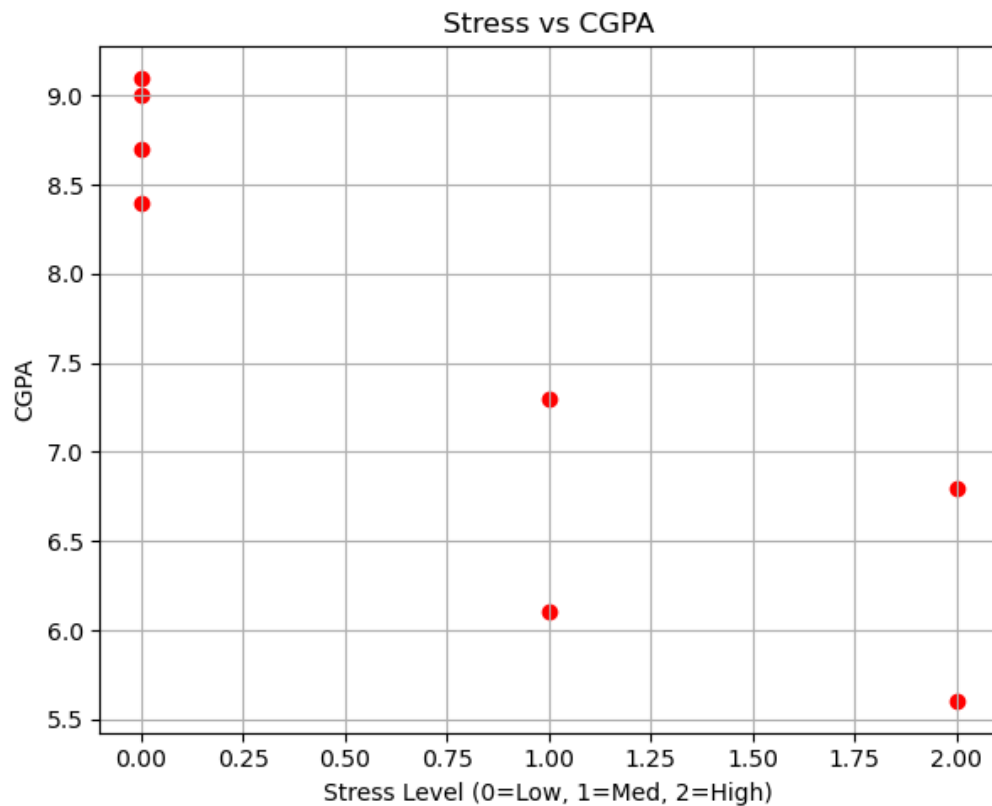
Final Correlation Value:

$$r = \frac{171.95}{174.64} = 0.985 \text{ (Very strong positive correlation)}$$

EDA – Correlations

r value	Meaning
+1.0	Perfect positive relationship
+0.7	Strong positive
0	No relationship
-0.7	Strong negative
-1.0	Perfect negative

EDA – Correlations



Student	Stress	CGPA
1	Low	8.4
2	Low	9.1
3	Med	7.3
4	High	5.6
5	High	6.8
6	Low	9.0
7	Med	6.1
8	Low	8.7

EDA – Feature Distributions

Feature distribution describes **how the values of a feature are spread across the dataset.**

It helps us understand:

- Typical values (centre)
- Spread/variability
- Skewness (left/right)
- Presence of outliers
- Whether transformation is needed before ML

EDA – Feature Distributions

Attendance (%) Distribution

Type: Numerical (0–100)

Typical pattern:

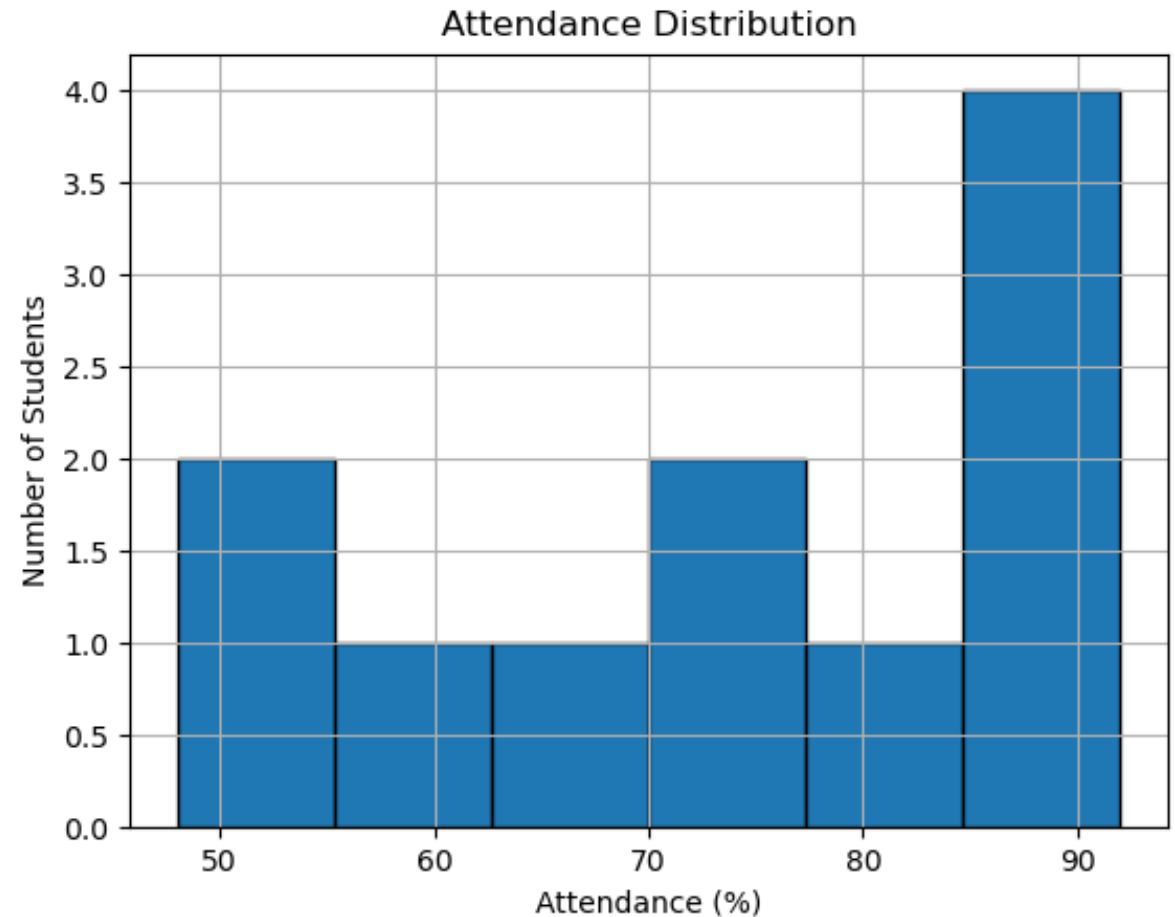
- Two clusters:
 - **45–60%** → At-risk students
 - **80–95%** → Regular students

Interpretation:

- Distribution is often **bimodal**
- Strong discriminator for dropout risk

EDA Insight:

- Students with attendance < 60% are significantly more likely to drop out.



EDA – Feature Distributions

CGPA Distribution

Type: Numerical (0–10)

Typical pattern:

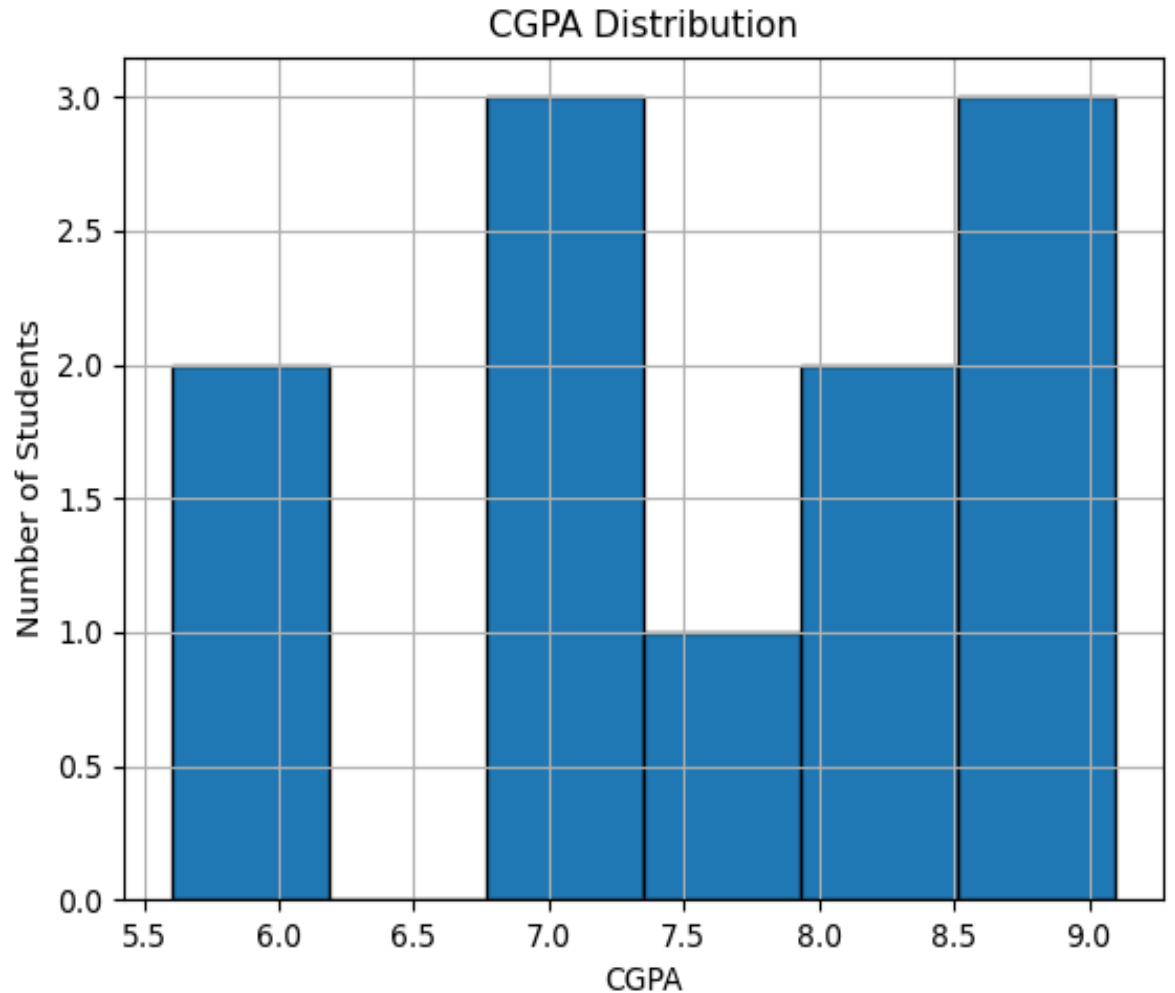
- Slightly **right-skewed**
- Majority between **6.5–8.5**
- Few extreme high performers (9–10)

Interpretation:

- Low CGPA tail (<6.5) signals academic risk

EDA Insight:

- CGPA is one of the strongest predictors of dropout.



EDA – Feature Distributions

LMS Login per Week Distribution

Type: Numerical (count)

Typical pattern:

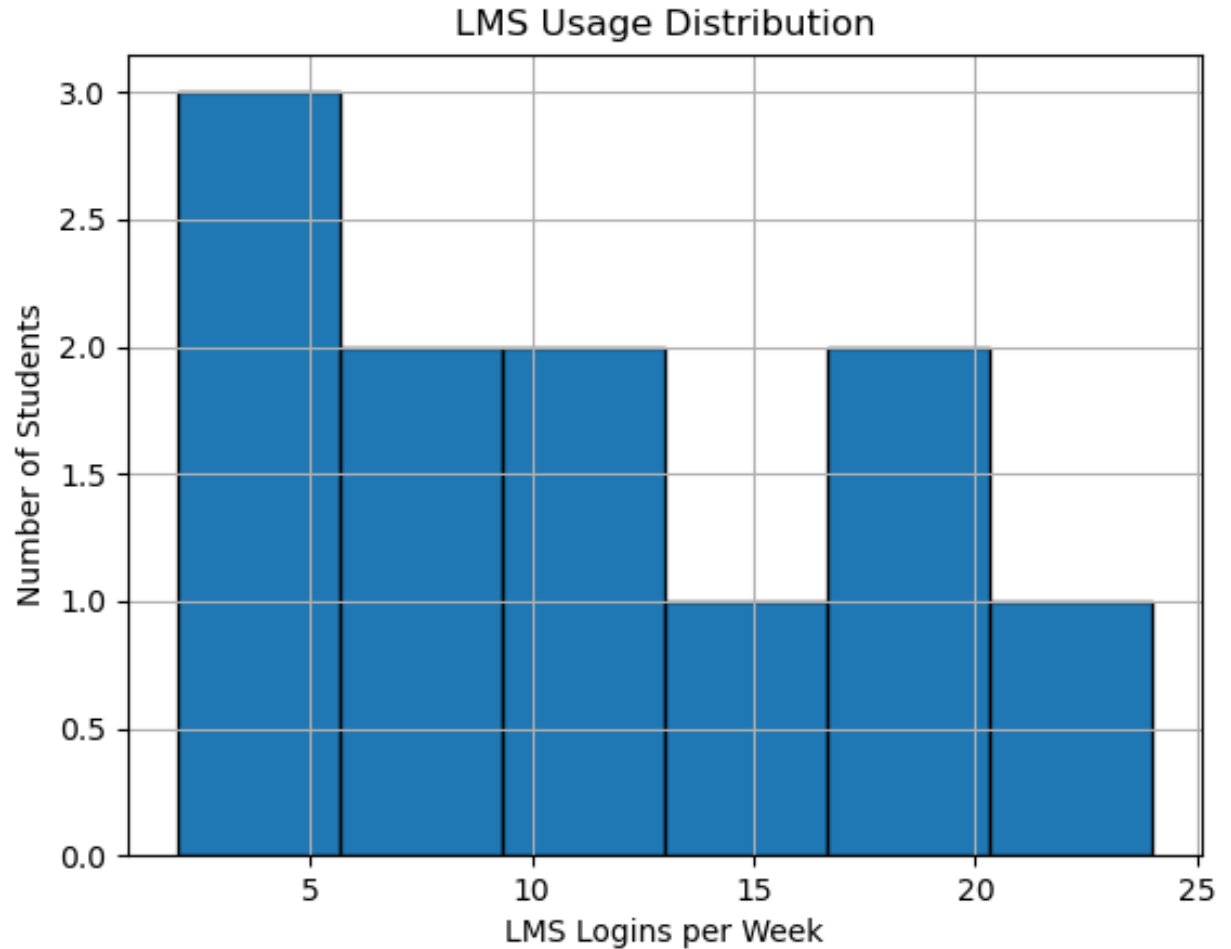
- **Right-skewed**
 - Many students with **0-5 logins**
 - Few students with **15-25 logins**

Interpretation:

- Large number of low-engagement students

EDA Insight:

- Low LMS usage indicates disengagement even if attendance is high.



EDA – Feature Distributions

WiFi Hours on Campus Distribution

Type: Numerical (hours)

Typical pattern:

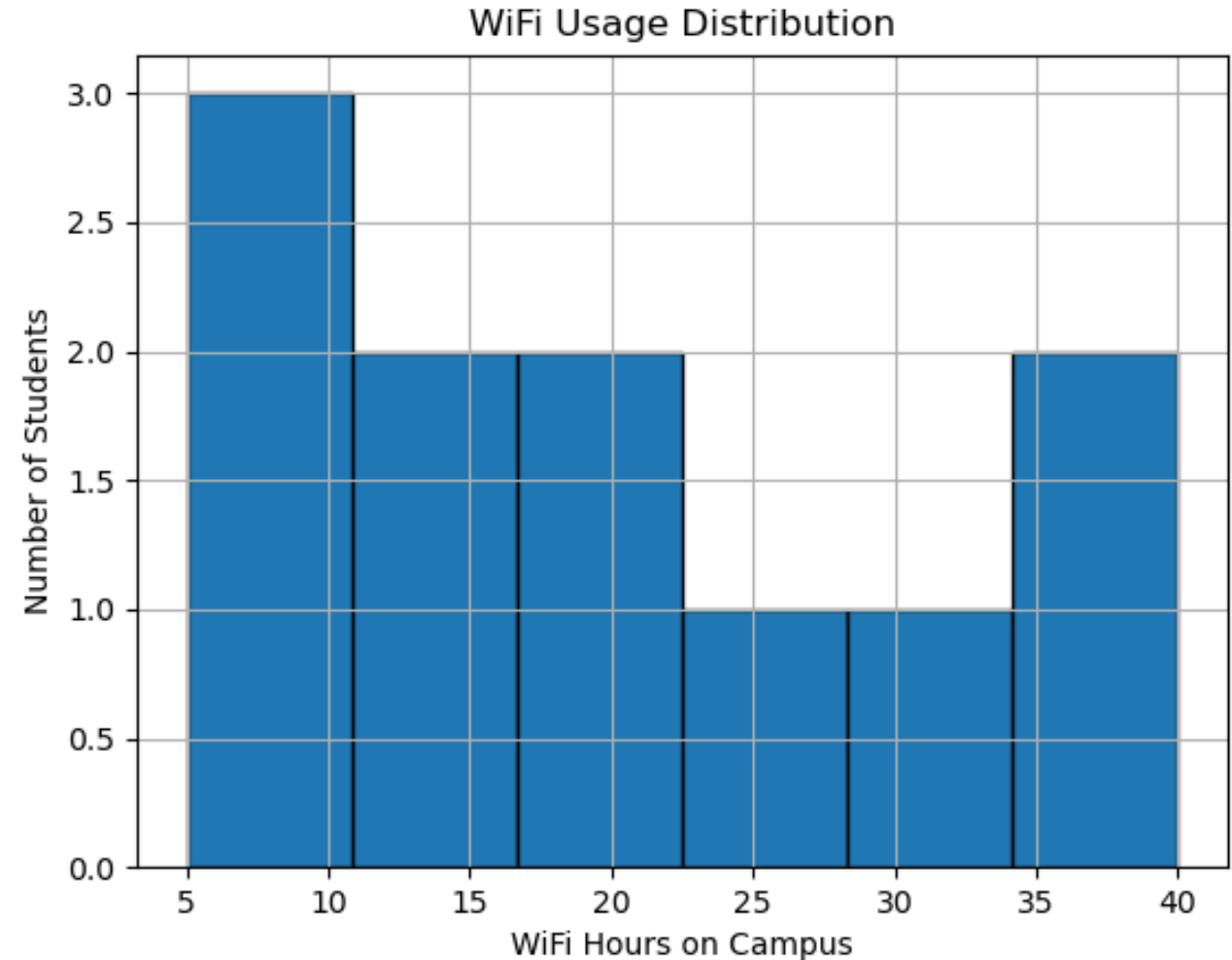
- Long-tailed distribution
- Some students spend very high time on campus

Interpretation:

- High WiFi hours + low LMS logins → non-academic campus presence

EDA Insight:

- WiFi hours alone are insufficient; must be combined with LMS data.



EDA – Feature Distributions

Stress Level Distribution

Type: Categorical (Low / Med / High)

Typical pattern:

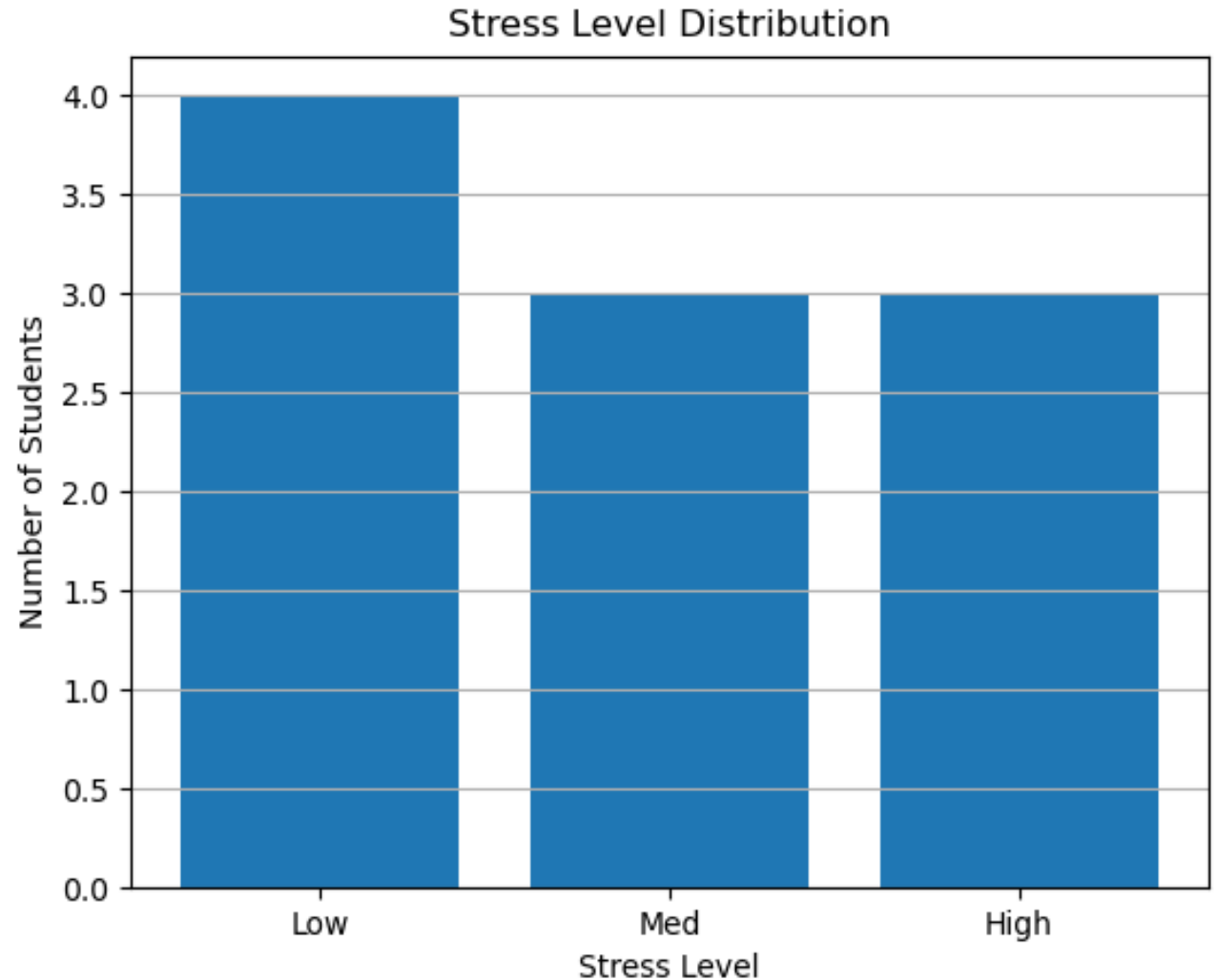
- Majority in **Medium**
- Smaller but important **High stress** group

Interpretation:

- High-stress students often overlap with low CGPA and engagement

EDA Insight:

- Stress is a strong negative factor affecting performance.



EDA – Feature Distributions

Mental Health Score Distribution

Type: Numerical (1–10)

Typical pattern:

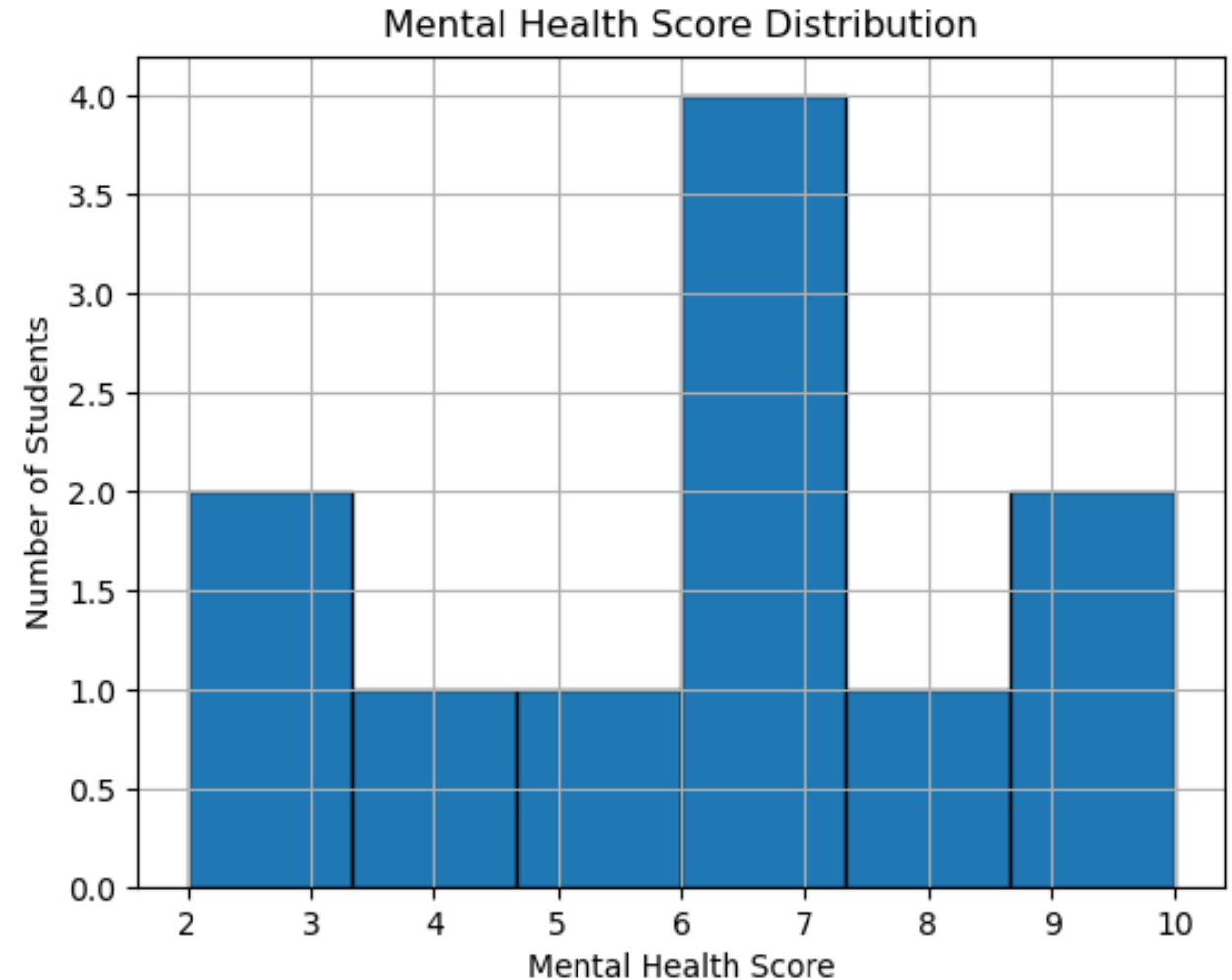
- Bell-shaped but slightly **left-skewed**
- Few students with very low scores (2–3)

Interpretation:

- Low mental health scores are early warning signs

EDA Insight:

- Mental health distribution highlights hidden dropout risk not visible in grades.



EDA – Feature Distributions

Engagement Score Distribution

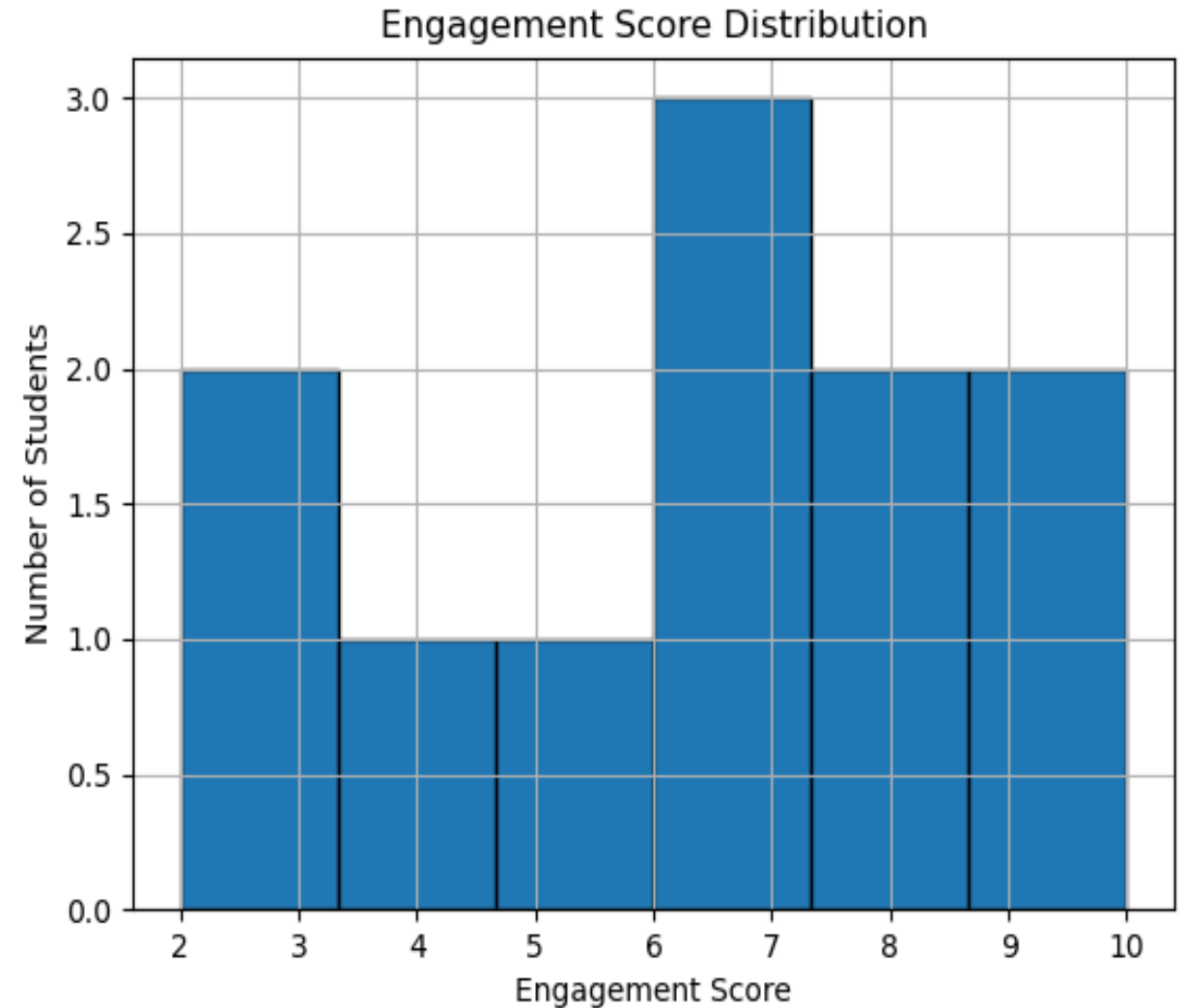
Type: Numerical (1–10)

Typical pattern:

- Closely mirrors mental health distribution
- Lower engagement cluster corresponds to high stress

EDA Insight:

- Engagement acts as a bridge between wellbeing and academic success.



EDA – Feature Distributions

Categorical Feature Distributions

Gender

- Often **imbalanced**
- Usually low predictive power

Department

- Often skewed toward one department
- Weak standalone predictor

Hostel / Fee Due

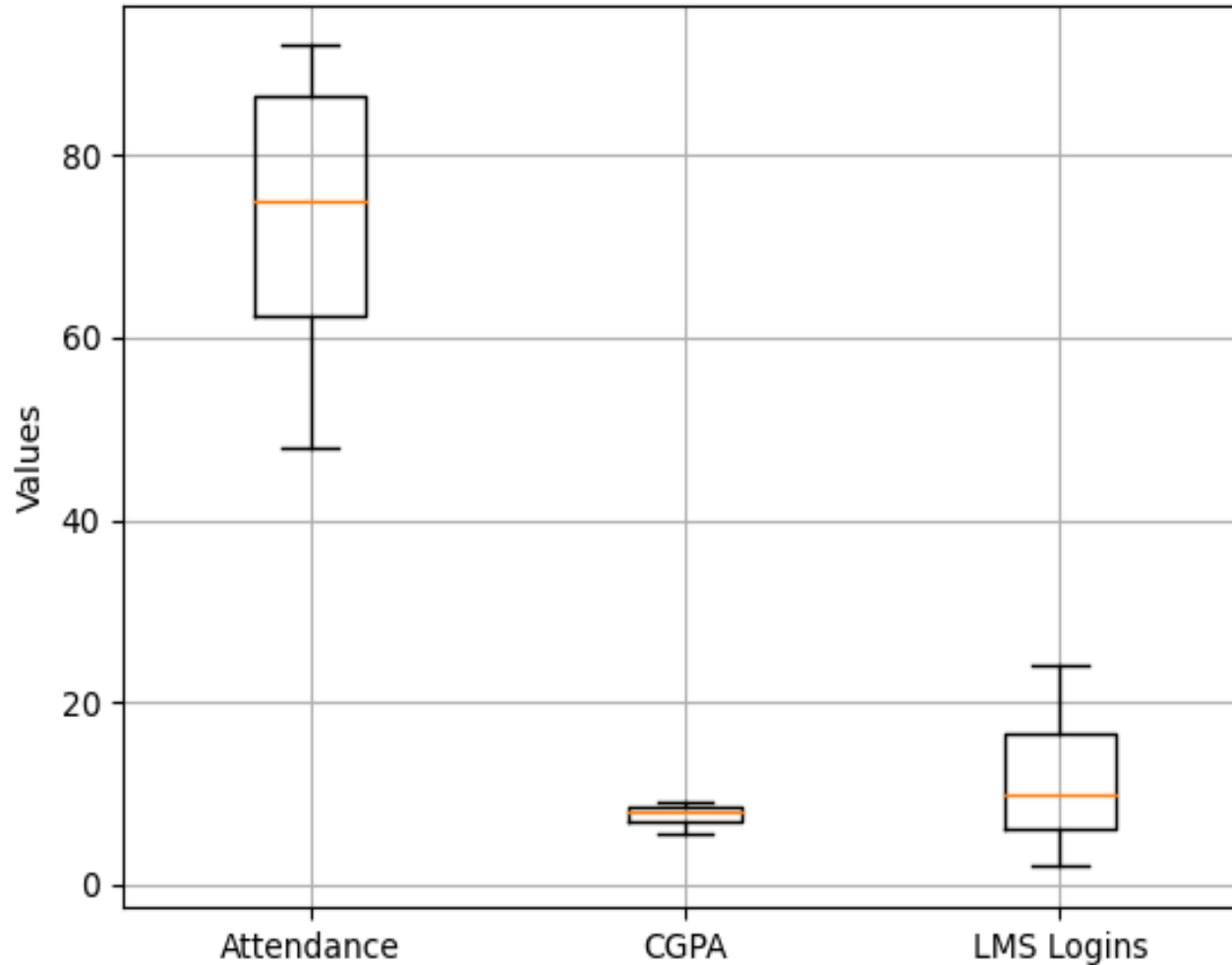
- Mostly binary with low variance

EDA Insight:

- Low-variance categorical features often add little value to ML models.

EDA – Feature Distributions

Boxplot of Key Features



- Attendance and CGPA distributions show clear separation between low- and high-performing students.
- LMS and WiFi distributions are right-skewed, indicating a small group of highly engaged students.
- Stress, mental health, and engagement distributions reveal hidden dropout risk beyond academic scores.

EDA – Feature Distributions

Feature	Distribution Type	Shape	EDA Insight
Attendance	Numerical	Bimodal	Strong dropout indicator
CGPA	Numerical	Slight right-skew	Academic risk marker
LMS Logins	Numerical	Right-skewed	Engagement indicator
WiFi Hours	Numerical	Long tail	Needs combination
Stress Level	Categorical	Imbalanced	Strong negative impact
Mental Health	Numerical	Left-skew	Early warning feature
Engagement	Numerical	Bell-shaped	Motivation indicator
Gender	Categorical	Imbalanced	Low importance

EDA - Takeaways

- Always **visualize distributions** before modelling
- Look for **skewness and outliers**
- Combine related features (e.g., WiFi + LMS)
- Remove **low-variance features**
- Use distributions to guide **feature engineering**

Feature Engineering

- Feature Engineering → **transform raw data into better inputs** for machine learning so models learn patterns more effectively.

Without feature engineering:

- Model sees **isolated values**

With feature engineering:

- Model understands **patterns**
- Early-warning signals appear **weeks earlier**
- Interventions become **actionable**

Feature Engineering

- Creating new features (derived variables)
- Encoding categorical
- Feature Normalizing / scaling
- Feature Reduction
- Interaction Features
- Final Engineered Feature Set

Creating new features (derived variables)

- Feature engineering formula is derived by **translating real-world behaviour into mathematics.**
- **FE → Encoding domain logic + data behaviour**
 - Step 1: Identify the Real-World Concept You Want to Capture
 - Step 2: Identify Relevant Raw Variables
 - Step 3: Decide the Mathematical Relationship
 - Step 4: Assign Direction of Influence
 - Step 5: Assign Weights (Relative Importance)
 - Step 6: Test with Hypothetical Scenarios

Creating new features (derived variables)

Step 1: Identify the Real-World Concept You Want to Capture

- What *latent behaviour* is not directly visible in raw columns?

Examples from your problem:

- “Academic struggle”
 - “True engagement”
 - “Burnout”
 - “Hidden risk”
- This is the **semantic goal** of the feature.

Creating new features (derived variables)

Step 2: Identify Relevant Raw Variables

- Select only variables that **directly contribute** to that concept.

Concept	Raw Features
Academic performance	CGPA, Arrears
Engagement	LMS Logins, WiFi Hours
Wellbeing	Stress, Mental Health, Engagement
Attendance effectiveness	Attendance, Arrears

Creating new features (derived variables)

Step 3: Decide the Mathematical Relationship

- Choose the operation based on **how variables interact**:

Relationship	Math Used	When to Use
Additive	+	Independent contributions
Penalizing	–	One factor reduces another
Multiplicative	×	Combined/compound effect
Weighted sum	$w_1x_1 + w_2x_2$	Unequal importance
Ratio	x / y	Trade-off relationships

Creating new features (derived variables)

Step 4: Assign Direction of Influence

- Determine: Does the feature **increase risk** or **reduce risk**?

Feature	Effect on Dropout
CGPA ↑	Risk ↓
Arrears ↑	Risk ↑
Stress ↑	Risk ↑
Engagement ↑	Risk ↓

Step 4: Assign Direction of Influence

Feature	Effect on Dropout Risk	Mathematical Action	Why
CGPA	CGPA $\uparrow \rightarrow$ Risk $\downarrow$	Invert or subtract	High CGPA is protective
Arrears_Count	Arrears $\uparrow \rightarrow$ Risk $\uparrow$	Add	More failures increase risk
Stress	Stress $\uparrow \rightarrow$ Risk $\uparrow$	Add	Higher stress worsens outcomes
Engagement	Engagement $\uparrow \rightarrow$ Risk $\downarrow$	Subtract or invert	Engagement protects against dropout
Attendance	Attendance $\uparrow \rightarrow$ Risk $\downarrow$	Subtract or invert	Regular attendance reduces risk
Mental Health	Mental health $\uparrow \rightarrow$ Risk $\downarrow$	Subtract	Emotional stability lowers risk
LMS Logins	LMS $\uparrow \rightarrow$ Risk $\downarrow$	Subtract	Active learning reduces risk
WiFi Hours	WiFi $\uparrow \rightarrow$ Slight risk $\downarrow$	Subtract (lower weight)	Indirect engagement

Creating new features (derived variables)

Step 5: Assign Weights (Relative Importance)

Weights are chosen using **one or more of the following**:

1. Domain Knowledge

- Arrears are more serious than small attendance drops
- LMS activity is more meaningful than WiFi presence

2. Statistical Evidence

- Correlation strength
- Feature importance from models

3. Interpretability Constraint

- Keep values easy to explain
- Avoid extreme coefficients

Weights usually:

- Sum to 1 (for averages)
- Or use small integers (1, 2, etc.)

Creating new features (derived variables)

Step 6: Test with Hypothetical Scenarios

A good formula must behave correctly in **edge cases**.

Ask:

- Does it increase risk when it should?
- Does it reduce risk for healthy students?
- If not → revise.

Creating new features (derived variables)

Example 1: Academic Risk Score

Step 1: Concept → Academic struggle

Step 2: Variables --> CGPA (↓ risk), Arrears (↑ risk)

Step 3: Relationship

- Penalize low CGPA
- Penalize arrears more strongly

Step 4: Direction

- Use $(10 - \text{CGPA})$
- Add arrears penalty

Step 5: Weighting

- Arrears more severe → multiply by 2

Final Formula

$$\text{AcademicRisk} = (10 - \text{CGPA}) + (2 \times \text{Arrears})$$

Creating new features (derived variables)

$$\text{Academic_Risk} = (10 - \text{CGPA}) + (2 \times \text{Arrears_Count})$$

Student	CGPA	Arrears	Academic Risk	Interpretation
A	8.8	0	$(10 - 8.8) + 0 = \mathbf{1.2}$	Low risk
B	7.0	1	$(10 - 7.0) + 2 = \mathbf{5.0}$	Medium risk
C	6.2	2	$(10 - 6.2) + 4 = \mathbf{7.8}$	High risk
D	5.5	3	$(10 - 5.5) + 6 = \mathbf{10.5}$	Very high risk

Creating new features (derived variables)

Example 2: Study Engagement Index

Step 1: Concept → True academic engagement

Step 2: Variables → LMS Logins (direct), WiFi Hours (indirect)

Step 3: Relationship → Weighted sum

Step 4: Direction → Both increase engagement

Step 5: Weighting

- LMS > WiFi → 0.6 vs 0.4

- LMS activity – Direct Engagement (Max weight)
- Wifi Hours – Indirect Engagement (Min weight)

Final Formula

$$\text{StudyEngagement} = (0.6 \times \text{LMS}) + (0.4 \times \text{WiFi})$$

Creating new features (derived variables)

$$\text{Study_Engagement} = (0.6 \times \text{LMS_Logins}) + (0.4 \times \text{WiFi_Hours})$$

LMS Logins	WiFi Hours	Study Engagement	Meaning
15	10	13.0	High academic engagement
3	25	12.2	Campus presence without studying
2	5	3.8	Very low engagement

Creating new features (derived variables)

Example 3: Wellbeing Score

Step 1: Concept → Emotional stability

Step 2: Variables → Mental Health, Engagement, Stress

Step 3: Relationship → Add positives, subtract negatives

Step 4: Direction

- Stress reduces wellbeing

Step 5: Weighting → Simple, interpretable subtraction

Final Formula

$$\text{Wellbeing} = \text{MentalHealth} + \text{Engagement} - \text{StressNum}$$

StressNum: Low=0, Med=1, High=2

Creating new features (derived variables)

$$\text{Wellbeing} = \text{MentalHealth} + \text{Engagement} - \text{StressNum}$$

Student	Mental Health	Engagement	Stress	Wellbeing	Interpretation
A	8	7	Low (0)	15	Emotionally stable
B	6	5	Med (1)	10	Moderate risk
C	4	3	High (2)	5	High dropout risk

Creating new features (derived variables)

Example 4: Attendance Quality

Step 1: Concept → Effective attendance

Step 2: Variables → Attendance, Arrears

Step 3: Relationship → Penalty

Step 4: Direction → Arrears reduce attendance value

Step 5: Weighting → -2 balances severity without overpowering attendance

Final Formula

$$\text{AttendanceQuality} = \text{Attendance} - (2 \times \text{Arrears})$$

Situation	Impact
Attendance drops from 85% → 83%	Minor
One arrear added	Major

Creating new features (derived variables)

Attendance %	Arrears	Attendance Quality	Meaning
90	0	90	Strong academic consistency
85	1	83	Minor risk
80	3	74	Hidden academic risk
75	4	67	High risk despite attendance

Encoding Categorical Features

- ML models require numeric inputs → mathematical computations

Feature	Raw	Engineered
Gender	M / F	0 / 1
Stress Level	Low / Med / High	0 / 1 / 2
Fee Due	Yes / No	1 / 0
Department	CSE, ECE, IT	0, 1, 2

Feature Scaling / Normalization

- Feature scaling (normalization) is the process of **rescaling numerical features to a common range**
- between **0 and 1**, so that all features contribute fairly to a machine learning model.

For a feature x with known maximum value:

$$x_{\text{scaled}} = \frac{x}{\max(x)}$$

This ensures:

$$0 \leq x_{\text{scaled}} \leq 1$$

Feature Scaling / Normalization

- Why Feature Scaling Is Necessary?

Consider two features:

- Attendance: **0 – 100**
- LMS Logins: **0 – 25**

In models that use:

- Distance (KNN, K-means)
- Gradients (Logistic Regression, Neural Networks)
- Regularization (L1/L2)
- **Large-range features dominate** the learning process.

Feature Scaling / Normalization

Before Scaling

Example:

- Attendance = 80
- LMS Logins = 10

Distance or weight contribution:

- Attendance contributes 80 units
- LMS contributes only 10 units
- The model incorrectly assumes **attendance is more important**, even if LMS usage is equally predictive.

After Scaling

- $\text{Attendance_scaled} = \frac{80}{100} = 0.80$
- $\text{LMS_scaled} = \frac{10}{25} = 0.40$

Distance or weight contribution:

- Attendance contributes 0.80 units
- LMS contributes only 0.40 units
- Now both features contribute **proportionately**.

Feature Scaling / Normalization

- Which Models Require Scaling Most?

Model	Computation	Scaling Needed?
KNN	Euclidean Distance	Yes
K-Means	Cluster Distance	Yes
Logistic Regression	Weighted sum + sigmoid	Yes
Neural Networks	Weighted sums	Yes
SVM	Hyper-plane Distance	Yes
Decision Trees	Threshold-based splitting (Value comparison)	Not mandatory
Random Forest	Threshold-based splitting (Value comparison)	Not mandatory

Feature Reduction (Remove Weak Features)

- Based on **variance, correlation and domain logic**.

Remove:

- Student_ID (identifier)
- Gender (low impact)
- Department (weak predictor)
- Fee_Due (low variance)

Keep:

- Attendance
- CGPA
- Arrears
- Stress
- Mental Health
- Engagement
- LMS Logins
- WiFi Hours

Interaction Features

- An interaction feature is created when the **effect of one feature depends on the value of another**.
- Step 1: Identify Non-Additivity
- Step 2: Choose Mathematical Operator Based on Behaviour
- Step 3: Map Common Interaction Patterns
 - Case 1: Both High = High Impact → Multiplication
 - Case 2: One Reduces the Other → Penalized Difference
 - Case 3: Only Dangerous When Combined → Multiplicative Penalty
 - Case 4: Efficiency or Balance → Ratio
 - Case 5: Unequal Importance → Weighted Interaction
- Step 4: Test Behaviour with Edge Cases

Interaction Features

Step 1: Identify Non-Additivity

- Does Feature A's impact change depending on Feature B?
- If **YES**, an interaction is needed.

Example:

- Low attendance **alone** → moderate risk
- Low CGPA **alone** → moderate risk
- **Low attendance + low CGPA together** → very high risk

Interaction Features

Step 2: Choose Mathematical Operator Based on Behaviour

- Operator Selection Rule

Real-World Interaction	Mathematical Mapping	Reason
Risk increases sharply only when both are high	Multiplication ($\times$)	Amplifies joint effect
One feature weakens another	Subtraction ($-$)	Penalization
Trade-off relationship	Ratio ($/$)	Balance
Unequal contribution	Weighted product/sum	Controlled influence

Interaction Features

Step 3: Map Common Interaction Patterns

Case 1: “Both High = High Impact” → Multiplication

Concept:

- Risk explodes when both features are bad

Formula:

$$\text{Interaction} = x \times y$$

Example:

$$\text{Academic_Consistency} = \text{Attendance} \times \text{CGPA}$$

- High $\times$ High $\rightarrow$ large
- Low $\times$ Low $\rightarrow$ very small
- Mixed $\rightarrow$ moderate
- This **cannot be captured by addition.**

Interaction Features

Case 2: “One Reduces the Other” → Penalized Difference

Concept:

- One feature reduces the effect of another

Formula:

$$x - y$$

Example:

$$\text{Wellbeing} = \text{MentalHealth} + \text{Engagement} - \text{Stress}$$

- Stress suppresses wellbeing even if engagement is high.

Interaction Features

Case 3: “Only Dangerous When Combined” → Multiplicative Penalty

Concept:

- Feature is risky only when paired with another

Formula:

$$x \times (1 - y)$$

Example:

$$\text{Burnout_Index} = \text{Stress} \times (10 - \text{Engagement})$$

- High stress + low engagement → large value
- High stress + high engagement → reduced impact

Interaction Features

Case 4: “Efficiency or Balance” → Ratio

Concept:

- Performance depends on balance

Formula:

$$\frac{x}{y + 1}$$

Example:

$$\text{Dist Engagement Ratio} = \frac{\text{Engagement}}{\text{Distance} + 1}$$

- Long distance weakens engagement effectiveness.

Interaction Features

Case 5: Unequal Importance → Weighted Interaction

Concept:

- One feature matters more

Formula:

$$w_1x + w_2y \text{ or } w(x \times y)$$

Example:

$$0.6 \times \text{LMS} + 0.4 \times \text{WiFi}$$

Interaction Features

Step 4: Test Behaviour with Edge Cases

- A correct interaction formula must behave logically:

Case	Expected Behaviour
Both high	High interaction
One high, one low	Moderate
Both low	Low
One irrelevant	Suppressed effect

Final Engineered Feature Set

Feature Category	Features
Academic	CGPA, Arrears, AttendanceQuality
Behavioural	LMS_Logins, StudyEngagement
Wellbeing	StressNum, MentalHealth, Wellbeing
Contextual	Distance_Engagement_Ratio
Interaction	Attendance × CGPA

Practice

Customer_ID	Age	Gender	City	Avg_Monthly_Spend	Visits_Per_Month	Last_Purchase_Days	Discount_Used	Membership_Years	Rating	Repeat_Purchase
C001	28	F	Chennai	4500	6	12	Yes	2.5	4.6	Yes
C002	35	M	Bangalore	12000	10	5	No	5	4.9	Yes
C003	22	F	Chennai		3	45	Yes	0.5	3.8	No
C004	41	M	Delhi	8000		20	Yes	4	4.2	Yes
C005	30	F	Mumbai	2500	2	90	No	1	2.9	No
C006	52	M	Delhi	15000	12	3	Yes	10	4.8	Yes
C007	26	F	Chennai	3000	4	60	No	1.2	3.5	No
C008	38	M	Bangalore	9000	8	15	Yes	3.5	4.4	Yes

Practice

PART A: Data Cleaning Questions

1. Identify columns with missing values. How many missing entries exist per column?
2. How would you handle missing Avg_Monthly_Spend? Should mean or median be used? Justify.
3. Verify whether Rating values fall within a valid range. What action would you take if a rating exceeds the range?
4. Convert Discount_Used and Gender into numerical form. Which encoding technique would you use and why?
5. Is Customer_ID useful beyond identification? How would you detect duplicate customer records?

Practice

PART B: Feature Selection Questions

1. Identify the target variable. Is this a classification or regression task?
2. Would City always improve prediction? Under what condition should it be removed?
3. Which features are likely correlated with Repeat_Purchase? Explain without computing correlation values.
4. Could Visits_Per_Month and Last_Purchase_Days convey similar information? Should both be retained?
5. Which 3 features would a marketing team find most useful? Why?

Practice

PART C: Feature Engineering Questions

1. Customer Engagement Score: Create a new feature using: Visits_Per_Month Membership_Years and Rating. Provide a formula.
2. Transform Last_Purchase_Days into a feature where higher = better engagement.
3. Convert Avg_Monthly_Spend into: Low / Medium / High spender. Define thresholds.
4. Create an interaction feature between: Discount_Used and Avg_Monthly_Spend. Explain the intuition.
5. Create Is_Loyal_Customer using Membership_Years. What threshold would you choose?